

Computer Vision, Lecture 2

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<https://mvp.in.tum.de/courses/vision/>

Outline for Today

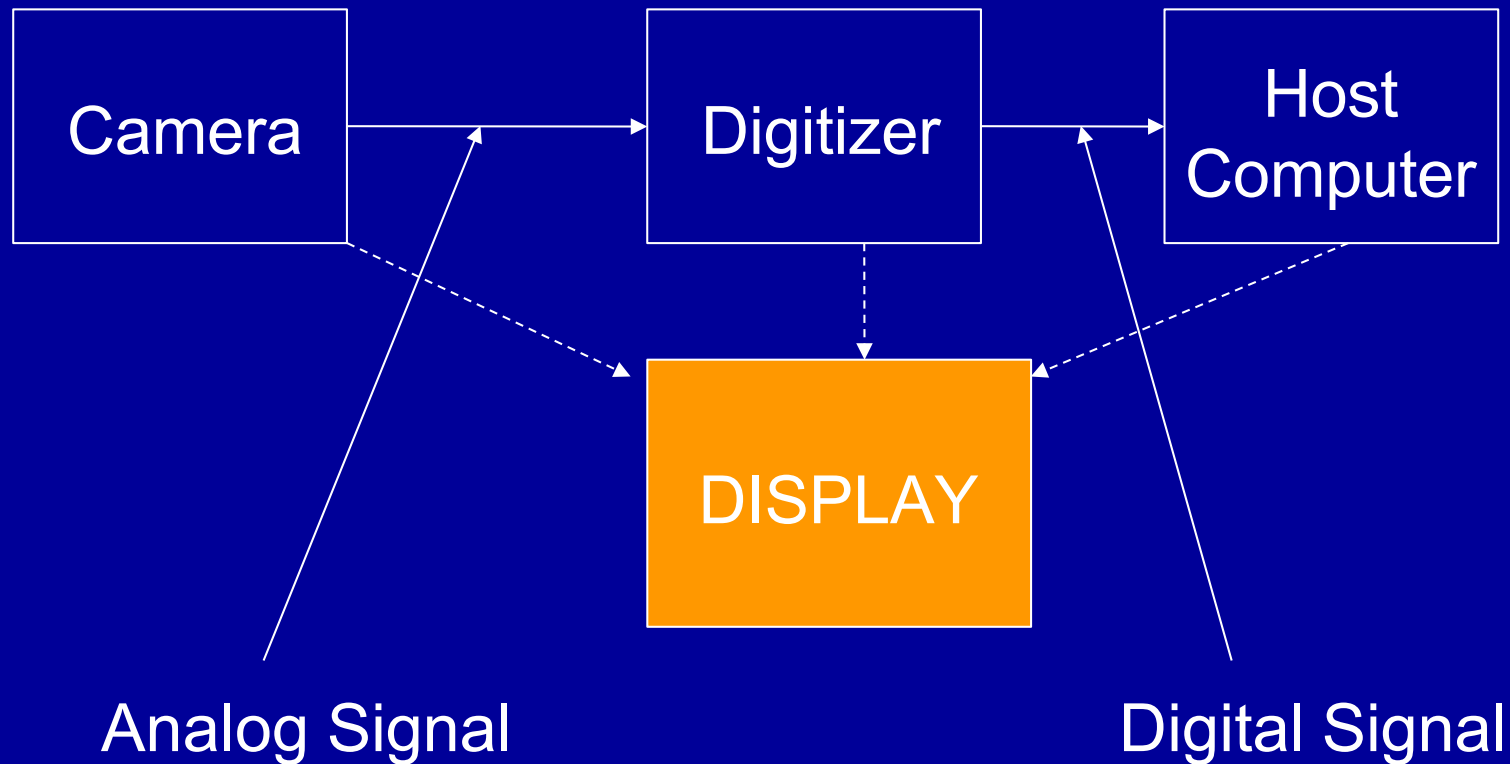
- What are images?
- Geometric Image Formation
- Camera Models

Cameras

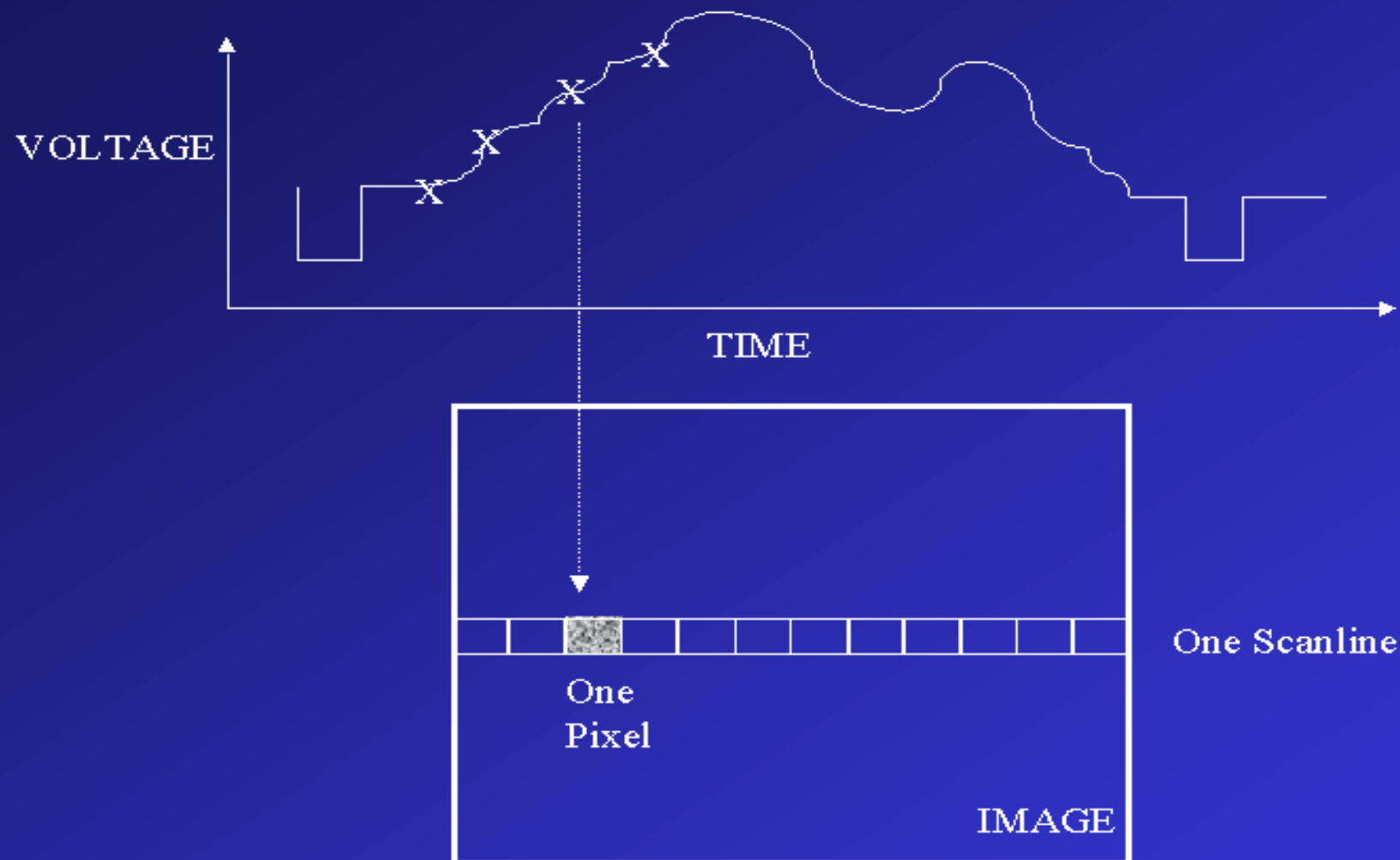
- First photograph due to Niepce,
- First on record shown - 1822
- Basic abstraction is the pinhole camera
 - lenses required to ensure image is not too dark
 - various other abstractions can be applied



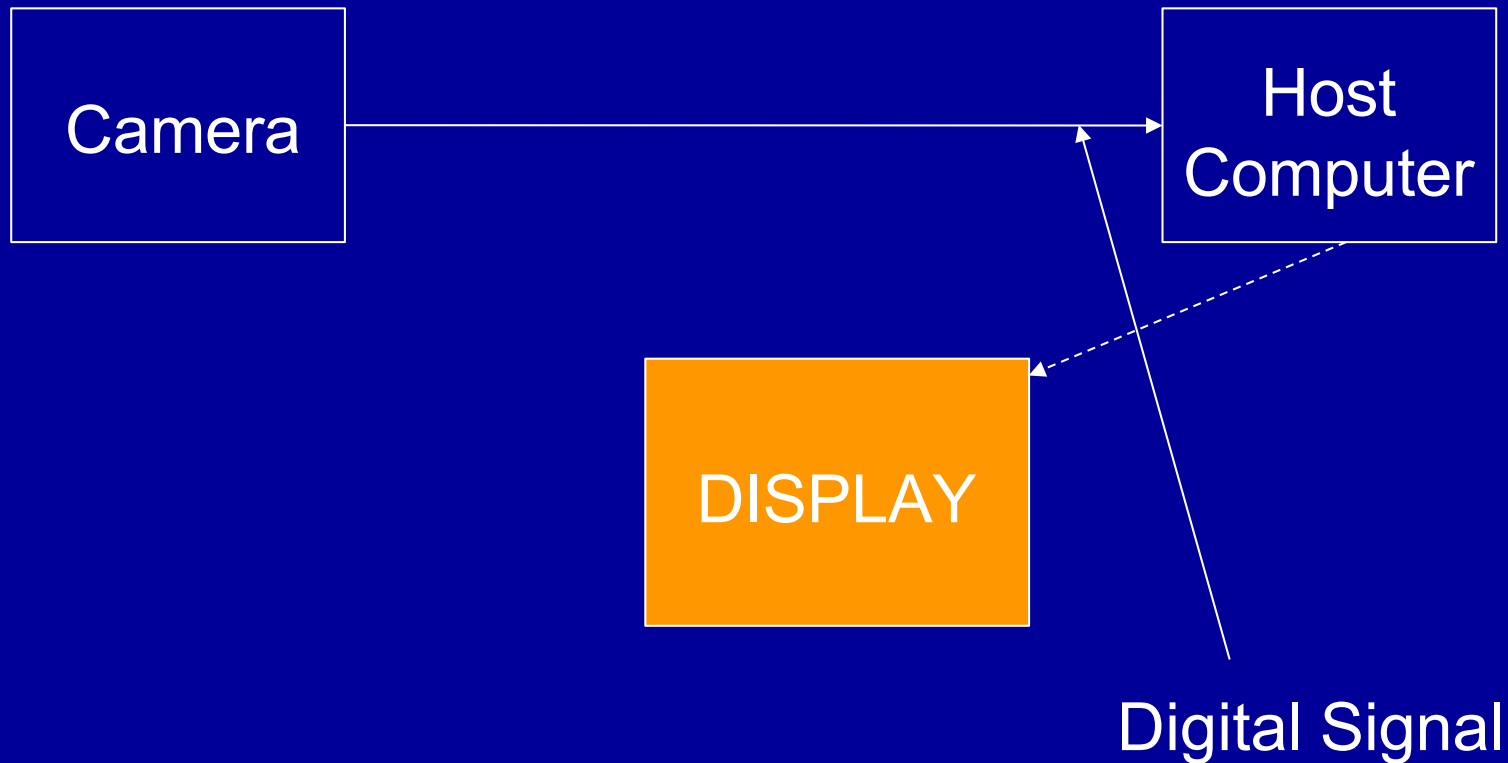
A Modern Digital Camera



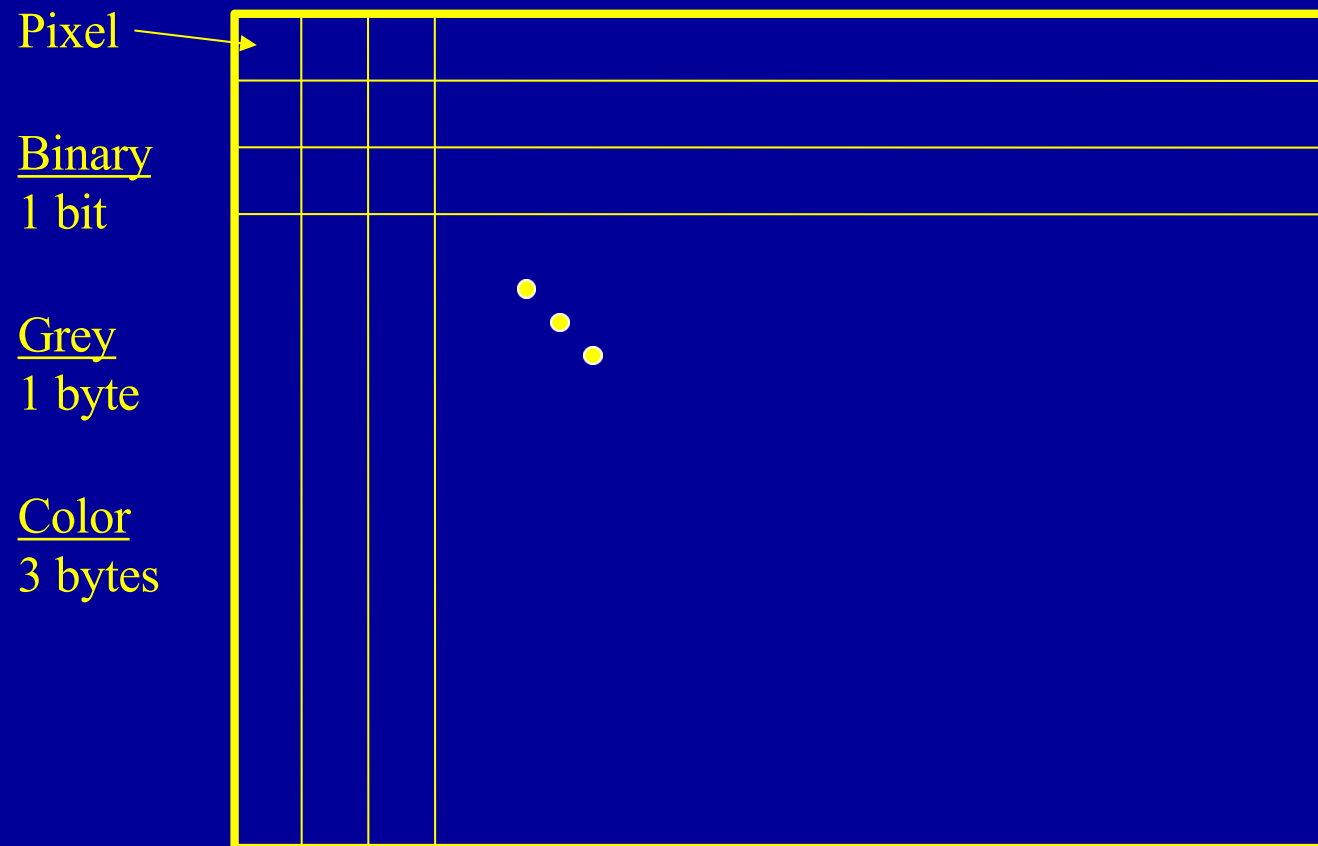
What's under the Hood



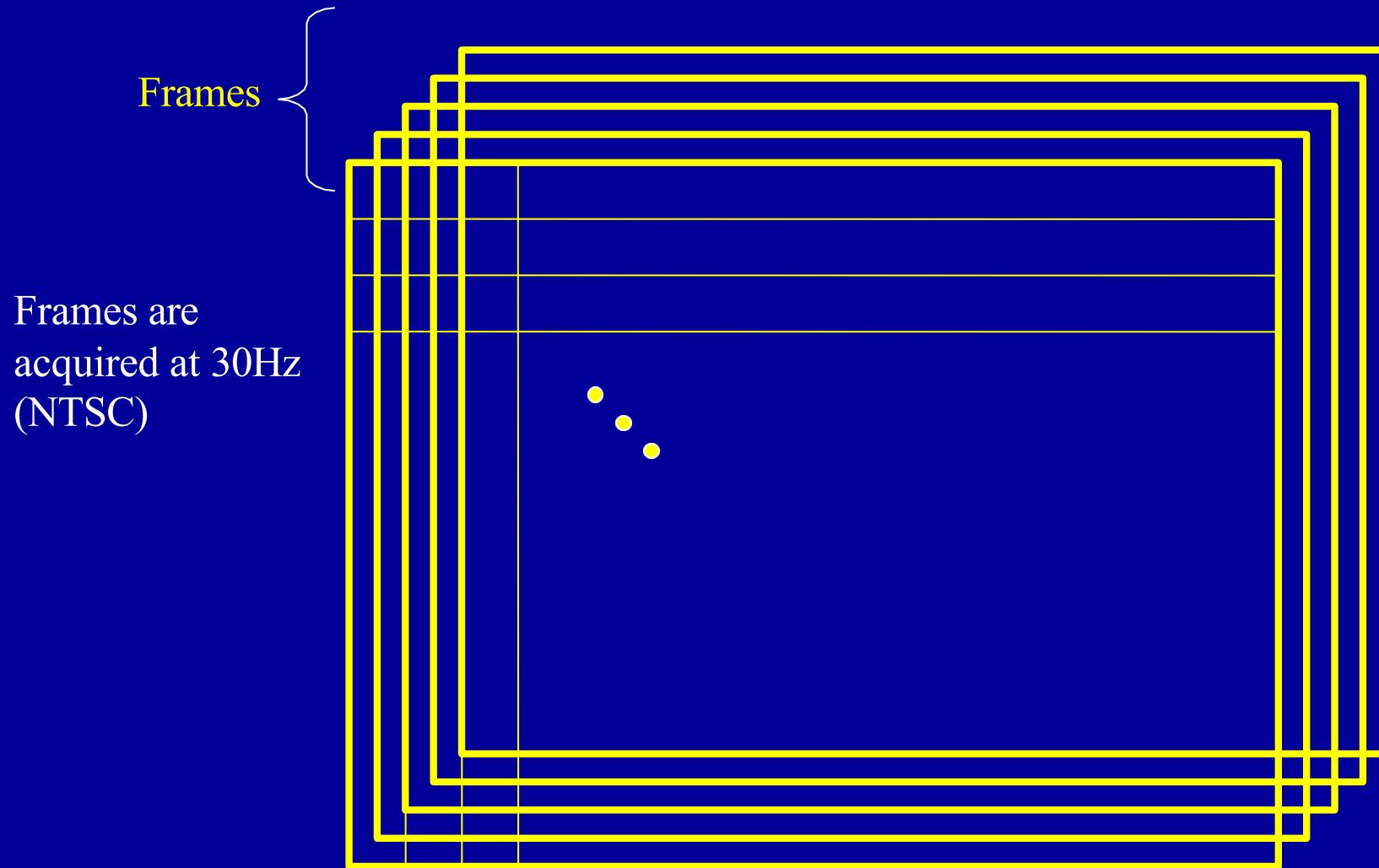
A Modern Digital Camera (Firewire)



THE ORGANIZATION OF A 2D IMAGE



THE ORGANIZATION OF AN IMAGE SEQUENCE



BANDWIDTH REQUIREMENTS

Binary

1 bit * 640x480 * 30 = 9.2 Mbits/second

Grey

1 byte * 640x480 * 30 = 9.2 Mbytes/second

Color

3 bytes * 640x480 * 30 = 27.6 Mbytes/second (actually about 37 mbytes/sec)

Typical operation: 3x3 convolution

9 multiplies + 9 adds \ 180 Mflops

Today's PC's are just getting to the point they
can process images at frame rate

Digitization Effects

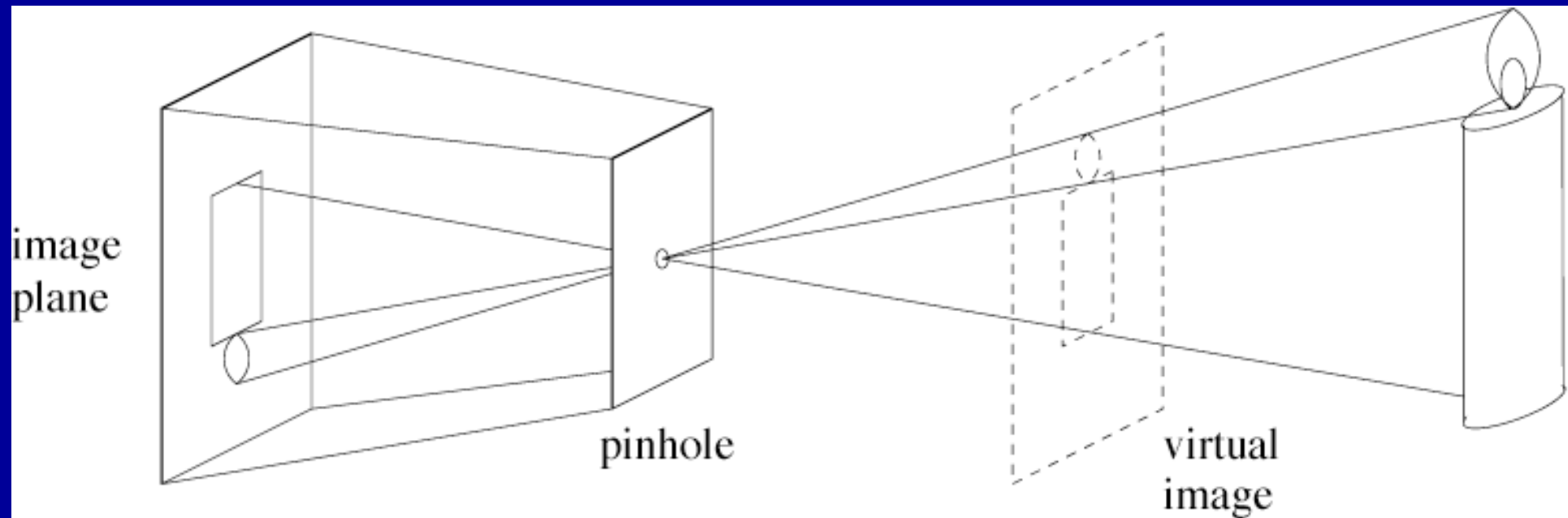
- The “diameter” d of a pixel determines the highest frequency representable in an image

$$d \leq 1 / 2l$$

- Real scenes may contain higher frequencies resulting in aliasing of the signal.
- In practice, this effect is often dominated by other digitization artifacts.

Pinhole cameras

- Abstract camera model - box with a small hole in it
- Pinhole cameras work in practice

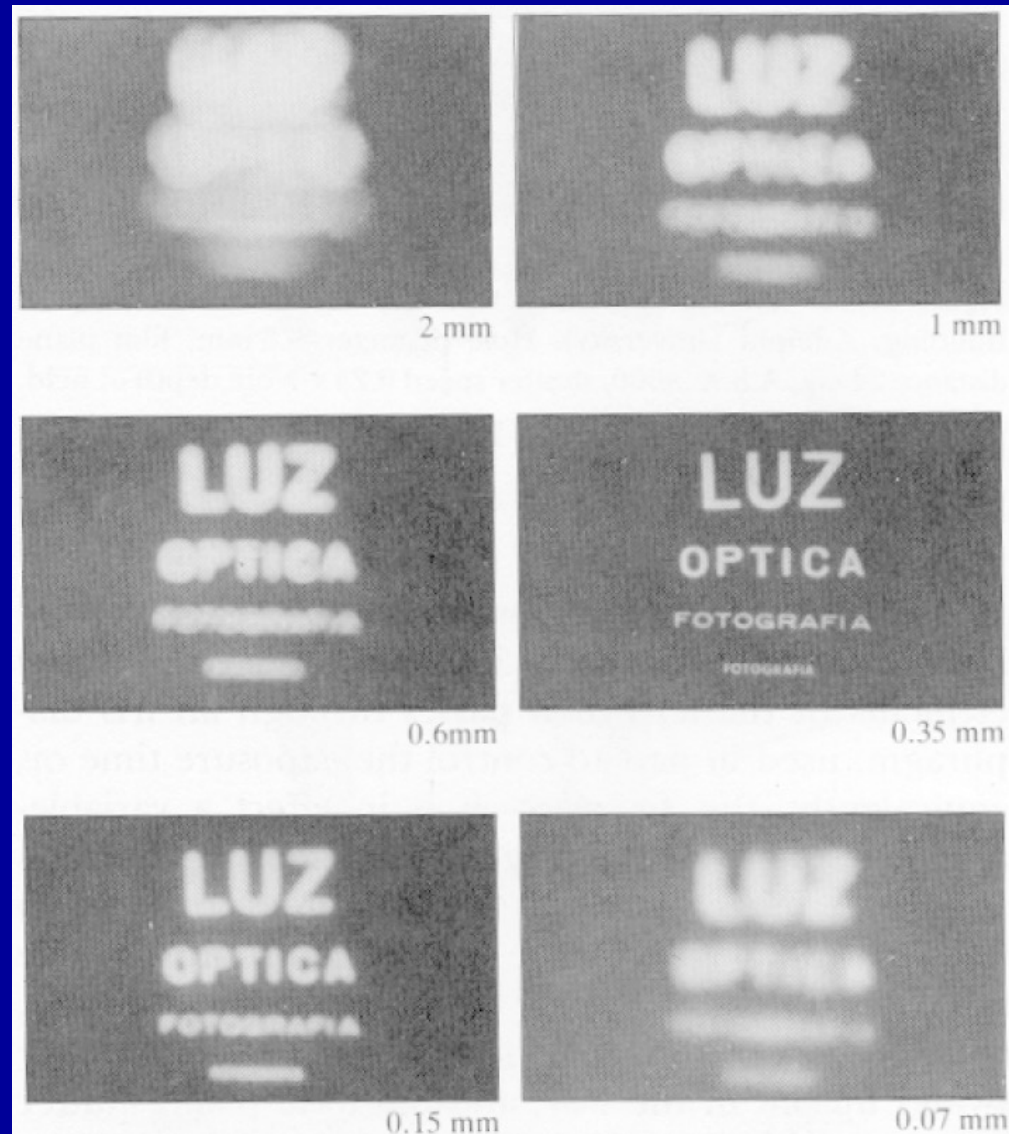


Real Pinhole Cameras

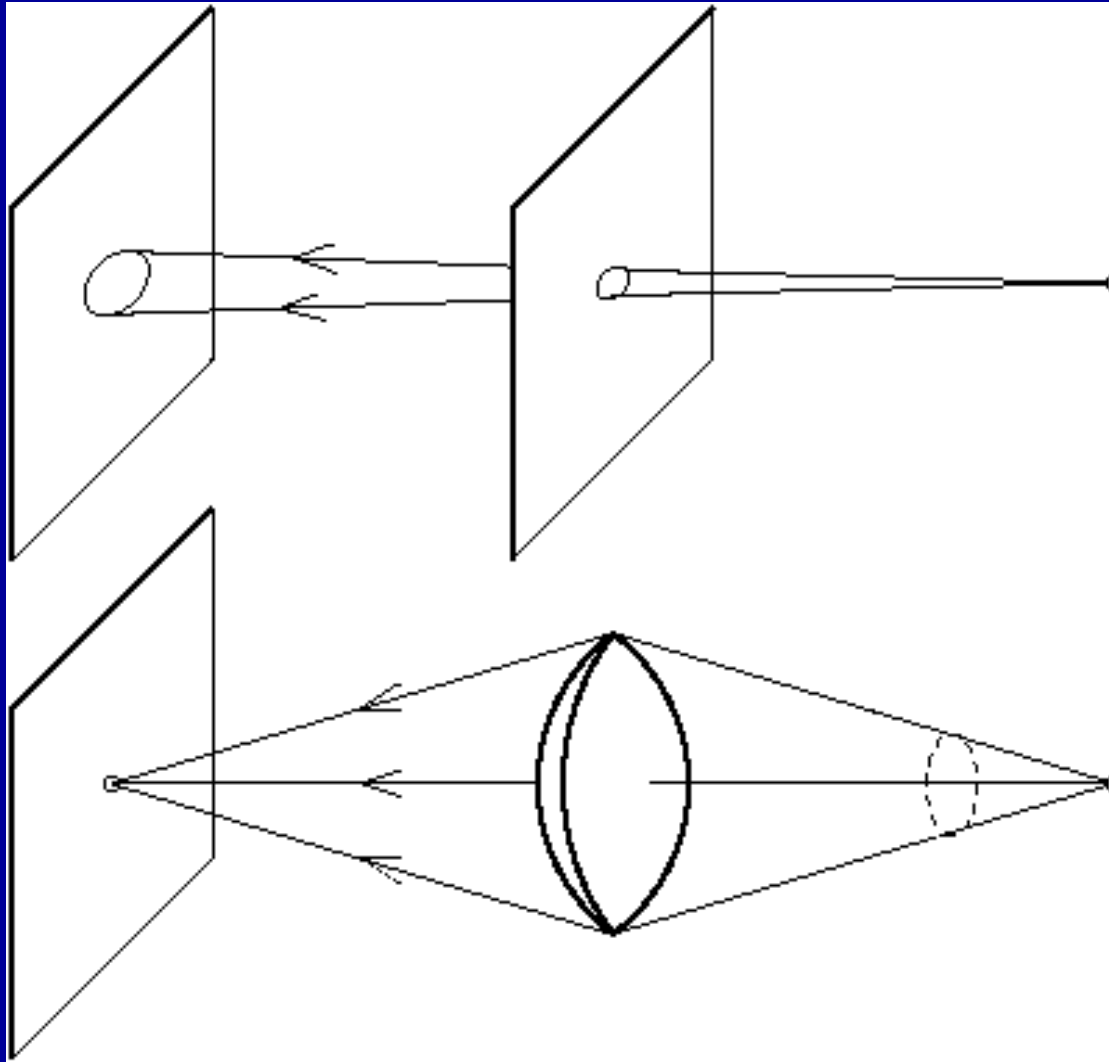
Pinhole too big -
many directions are
averaged, blurring the
image

Pinhole too small-
diffraction effects blur
the image

Generally, pinhole
cameras are *dark*, because
a very small set of rays
from a particular point
hits the screen.

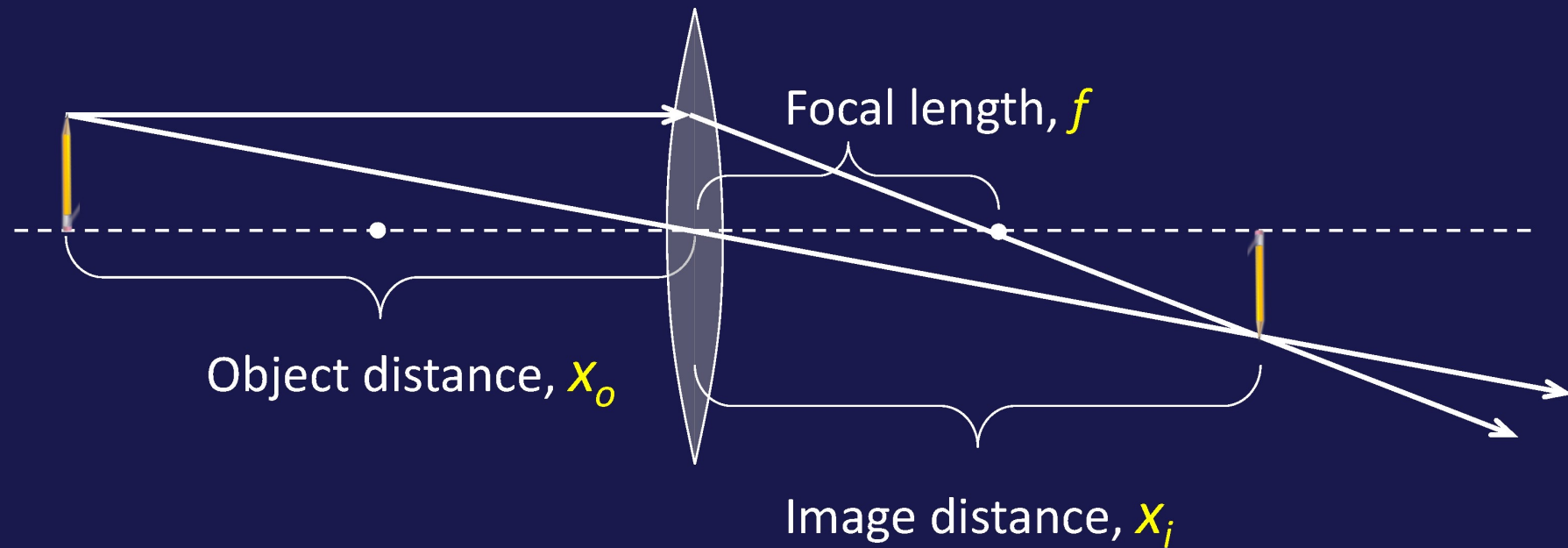


The reason for lenses



Lenses gather and focus light, allowing for brighter images.

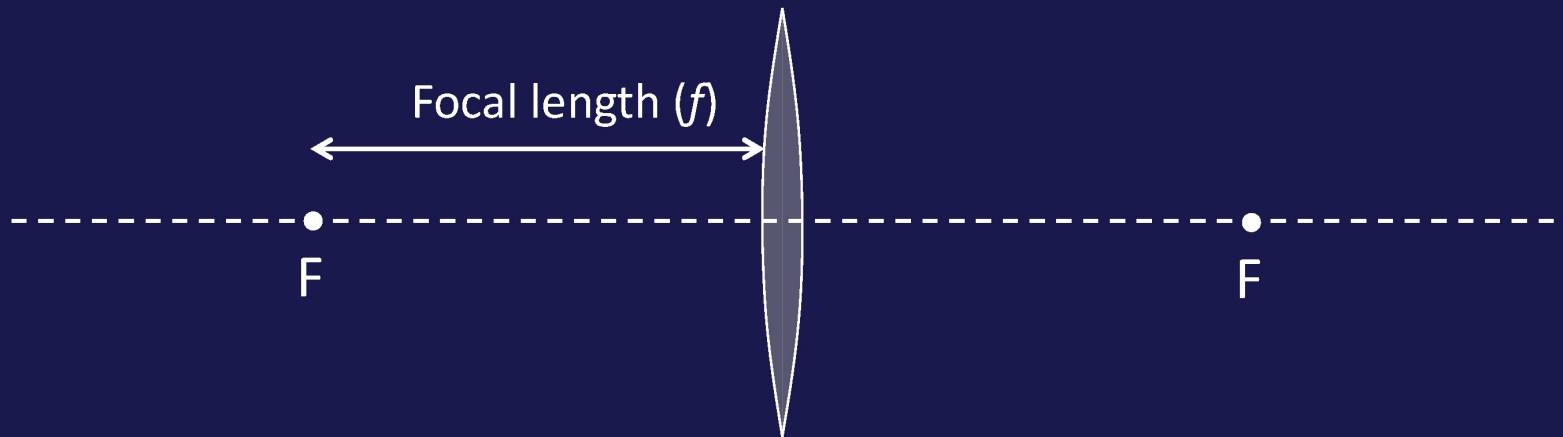
The Lens Equation



$$\frac{1}{x_o} + \frac{1}{x_i} = \frac{1}{f}$$

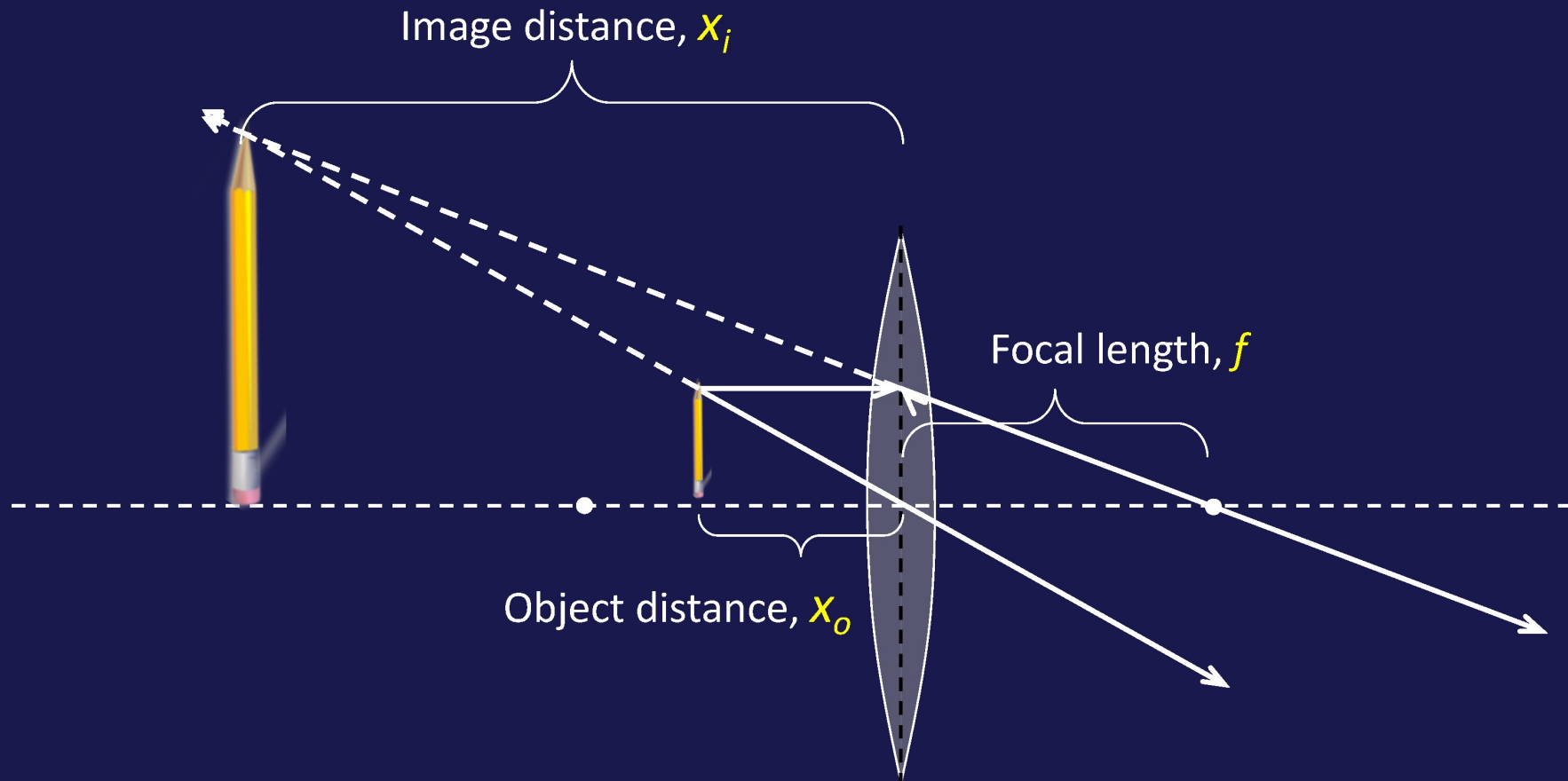
Focal Length

- Remember we defined the **focal length** for a lens



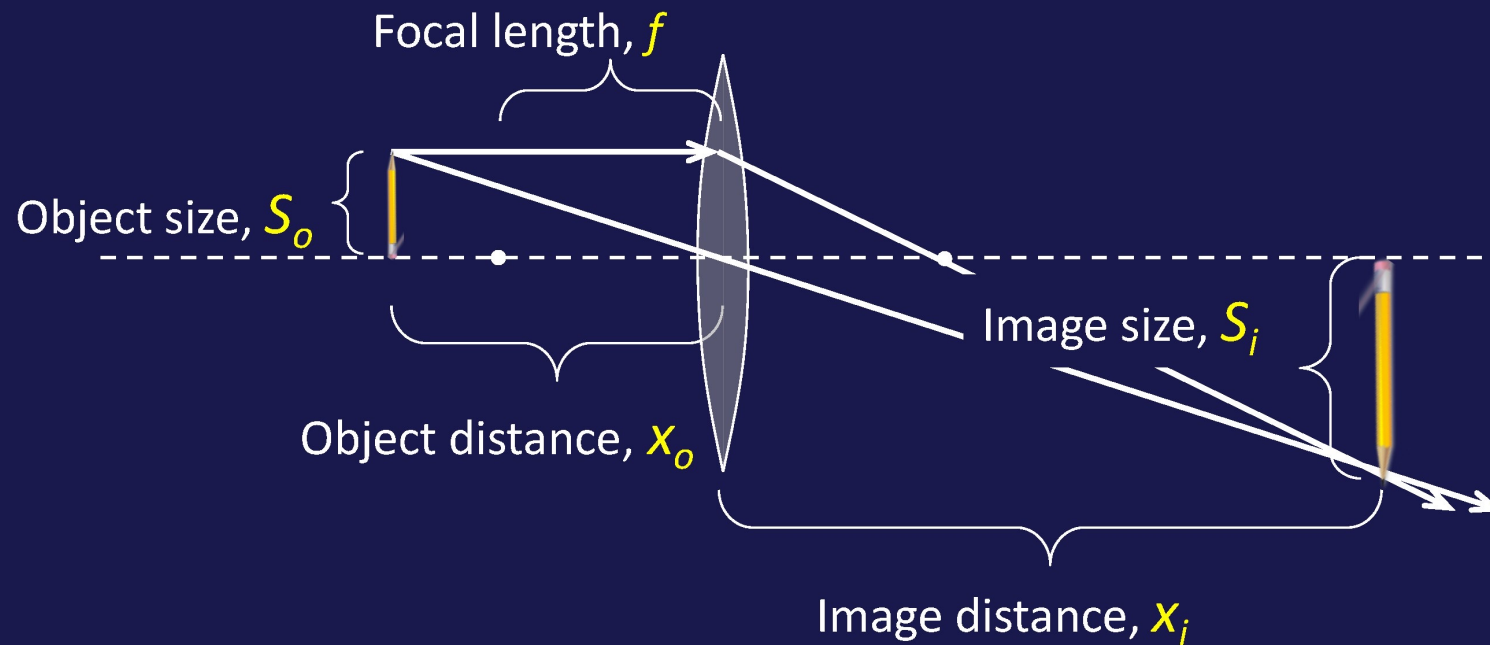
- We also defined the sign of f . The focal length, f , is defined as **positive** for converging lenses and **negative** for diverging lenses.

Lens Equation Quantities



- The image distance is **negative** for a (virtual) image on the left of the lens.

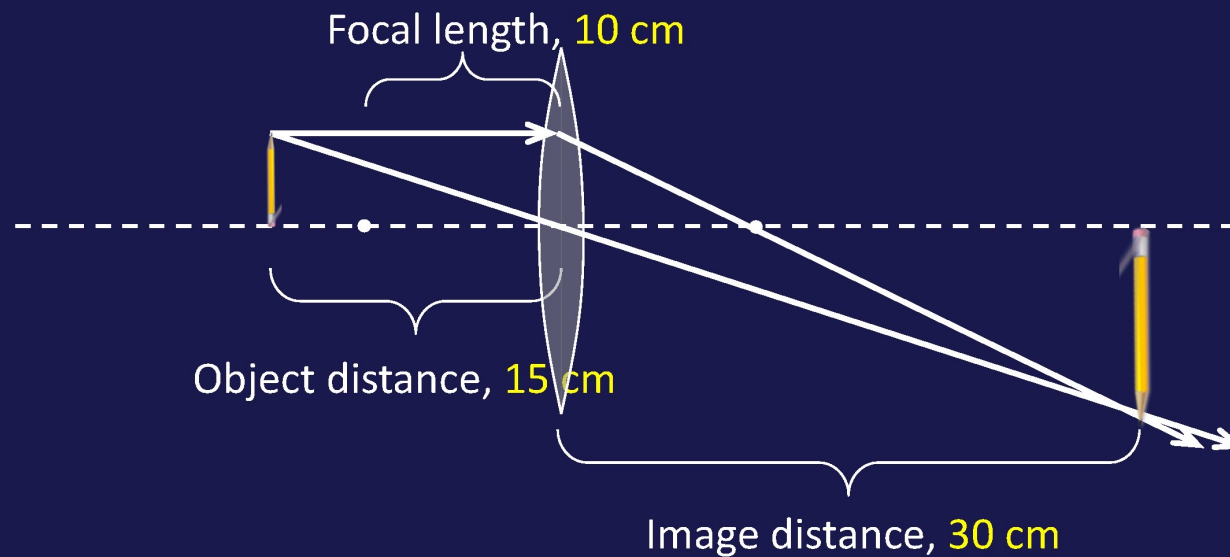
Magnification



$$M = \frac{S_i}{S_o} = -\frac{x_i}{x_o}$$

Magnification is indicated by the letter M, and is negative for an image inverted with respect to the object, and positive for an image that is not inverted

Magnification

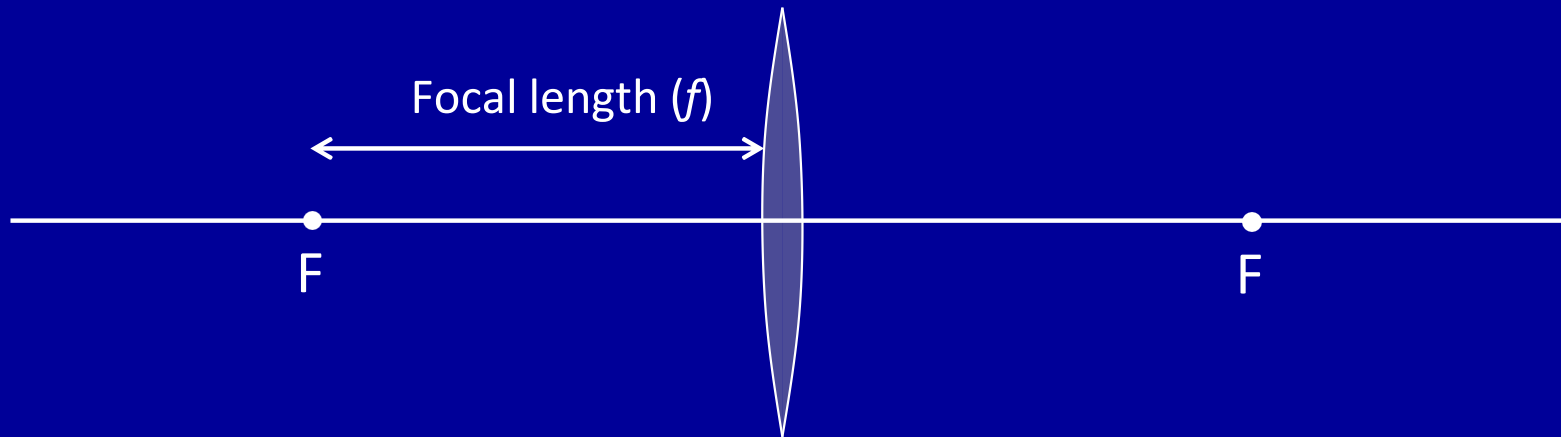


$$M = -\frac{x_i}{x_o} = -\frac{30 \text{ cm}}{15 \text{ cm}} = -2$$

Magnification of -2 means the image is twice the size of the object and inverted.

Focal Length

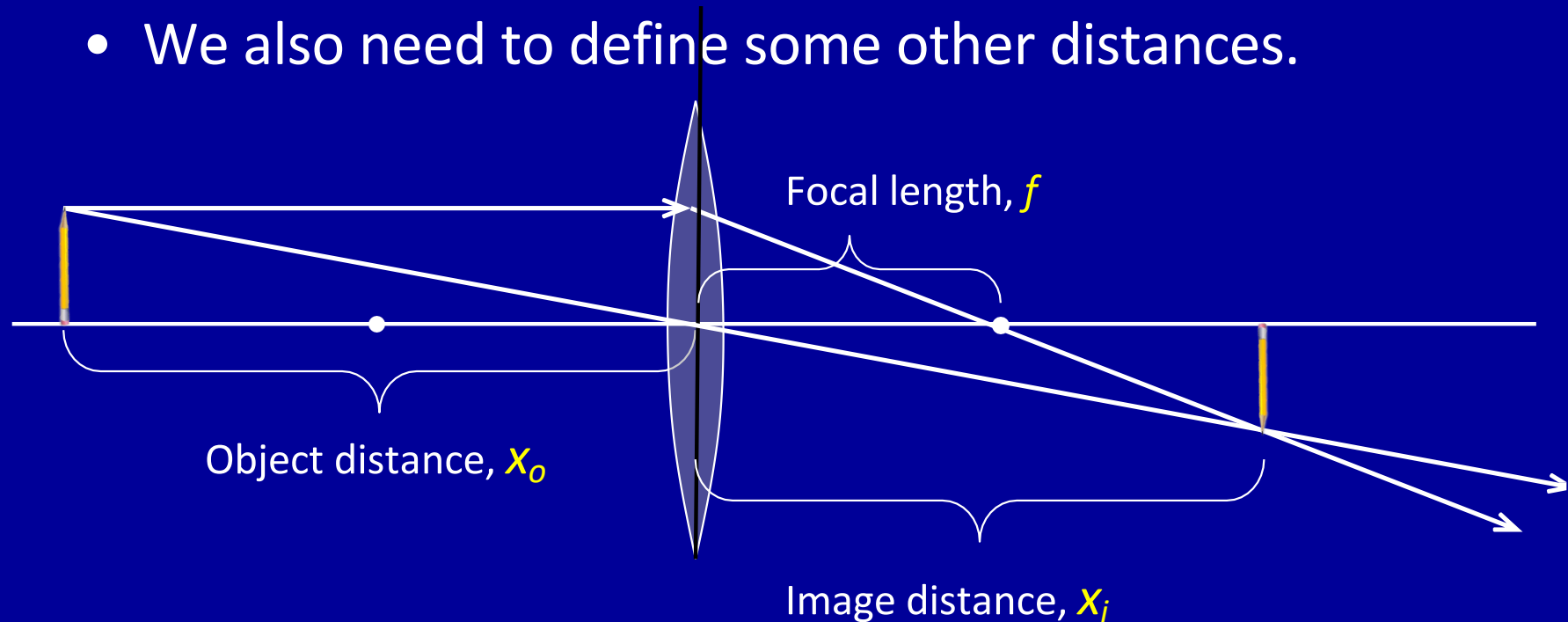
- Remember we defined the **focal length** for a lens



- We also defined the sign of f . The focal length, f , is defined as **positive** for converging lenses and **negative** for diverging lenses.

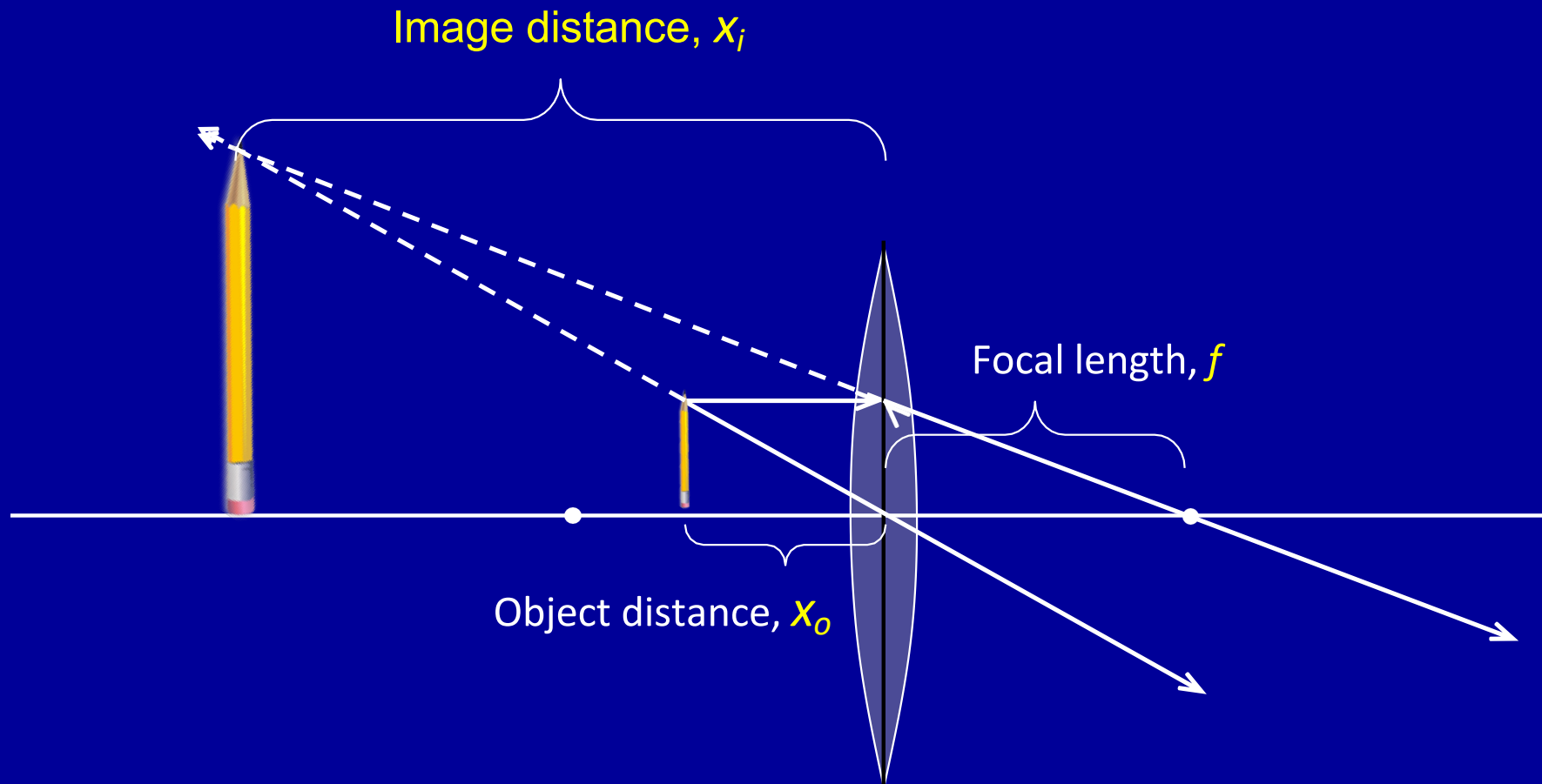
Lens Equation Quantities

- We also need to define some other distances.



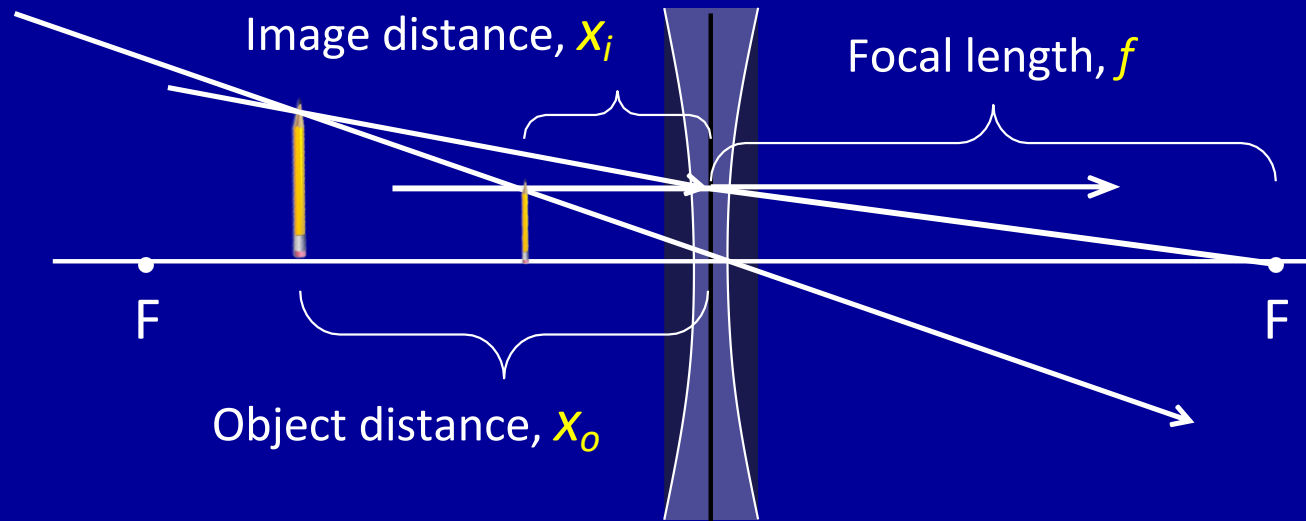
- The object distance is **positive** for an object to the left of the lens. The image distance is **positive** for a (real) image on the right of the lens. These quantities are negative for the reverse situation. Be careful with this.

Lens Equation Quantities



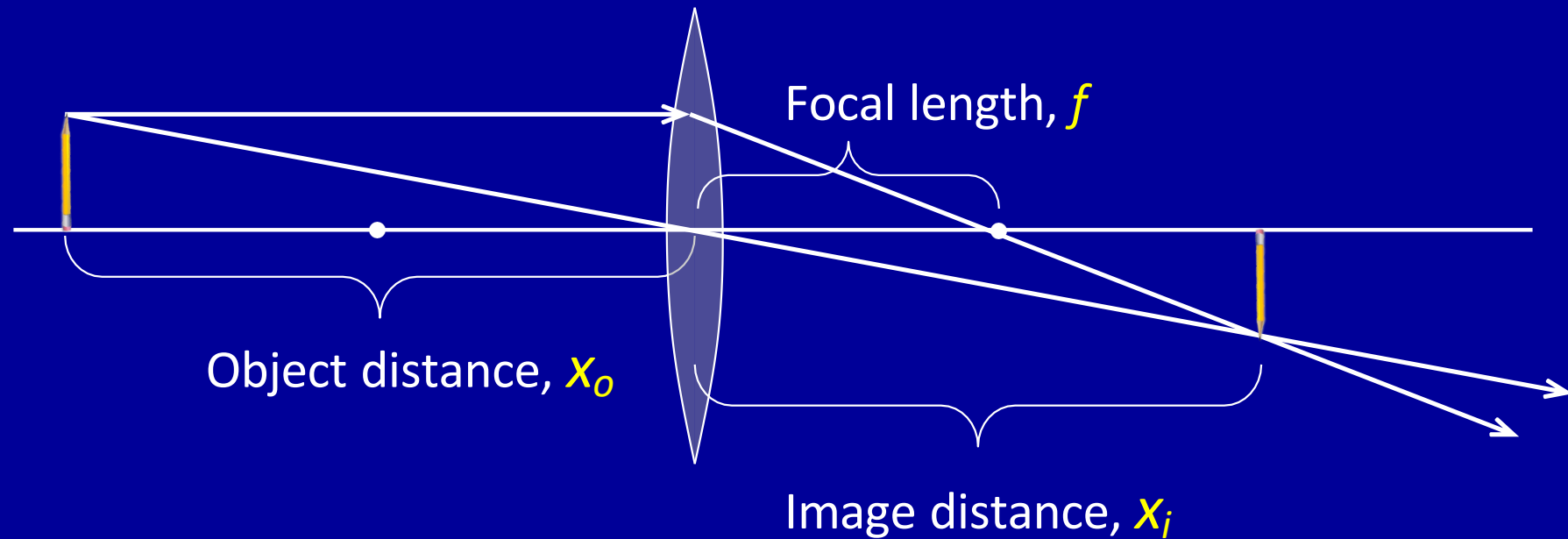
- The image distance is **negative** for a (virtual) image on the left of the lens.

Clicker Question



- Which quantities are **negative** in this example?
 - A. Image distance
 - B. Focal length
 - C. Object distance
 - D. A and B**
 - E. A and C

The Lens Equation



$$\frac{1}{x_o} + \frac{1}{x_i} = \frac{1}{f}$$

The Lens Equation

- **Given:**

- $f = 10 \text{ cm}$
- Object is 15 cm in front of lens: $x_o = 15$

- **Find:**

- Where is image and is it real or virtual?

- **Solve** equation for x_i :

- Substitute numbers for letters
- Subtract $1/15$ from both sides
- Arithmetic on calculator
- Multiply by $x_i/0.033$

$$\frac{1}{x_o} + \frac{1}{x_i} = \frac{1}{f}$$

$$\frac{1}{15} + \frac{1}{x_i} = \frac{1}{10}$$

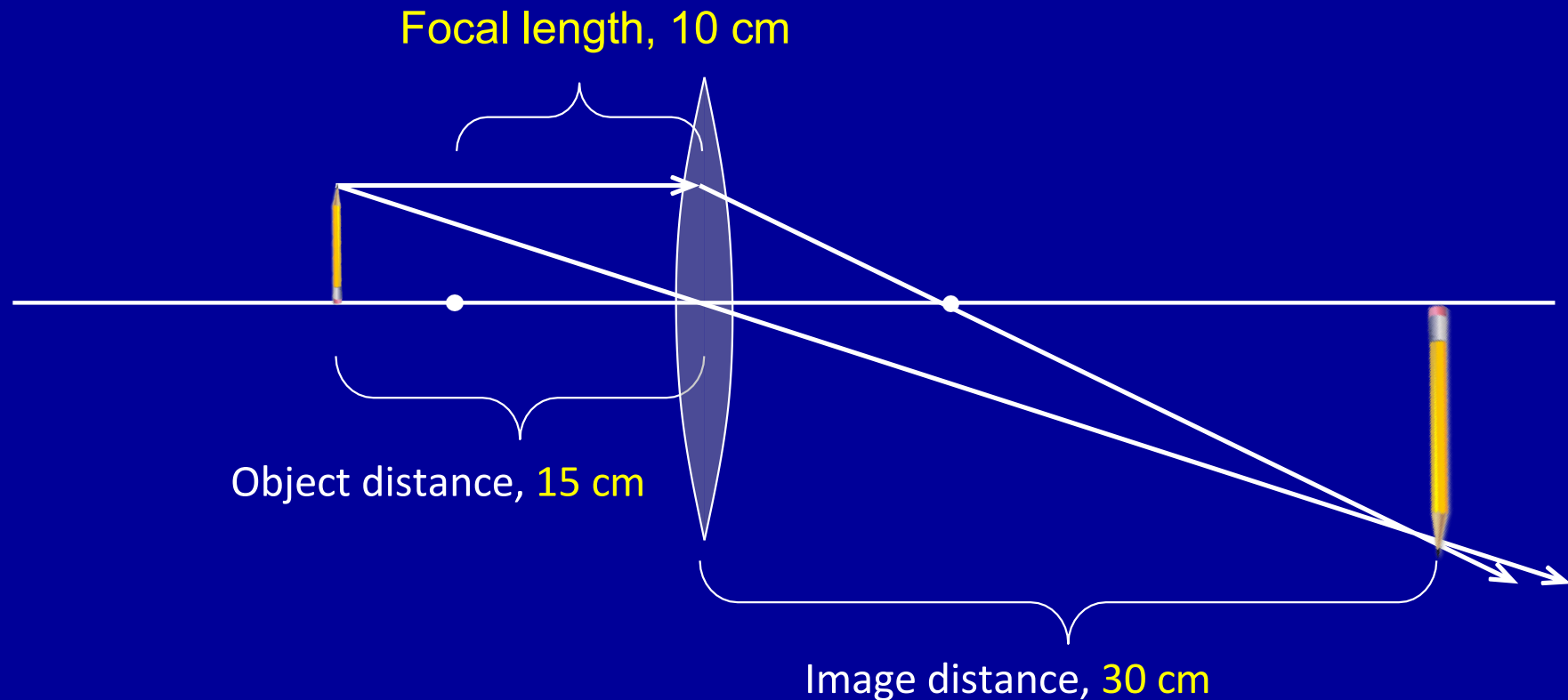
$$\frac{1}{x_i} = \frac{1}{10} - \frac{1}{15}$$

$$\frac{1}{x_i} = 0.033$$

$$x_i = \frac{1}{0.033} = 30 \text{ cm}$$

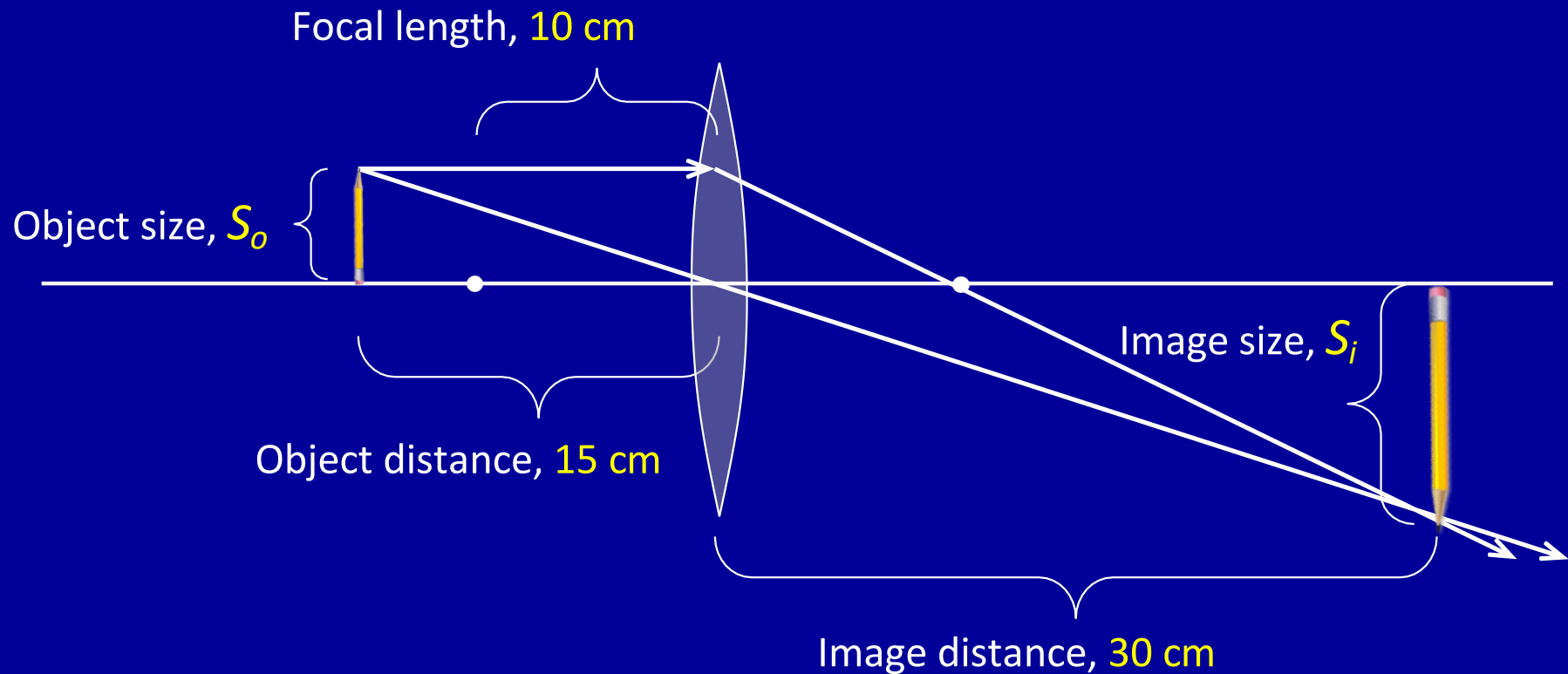
Image is 30 cm to the **right** of the lens and **real** because x_i is positive

The Lens Equation



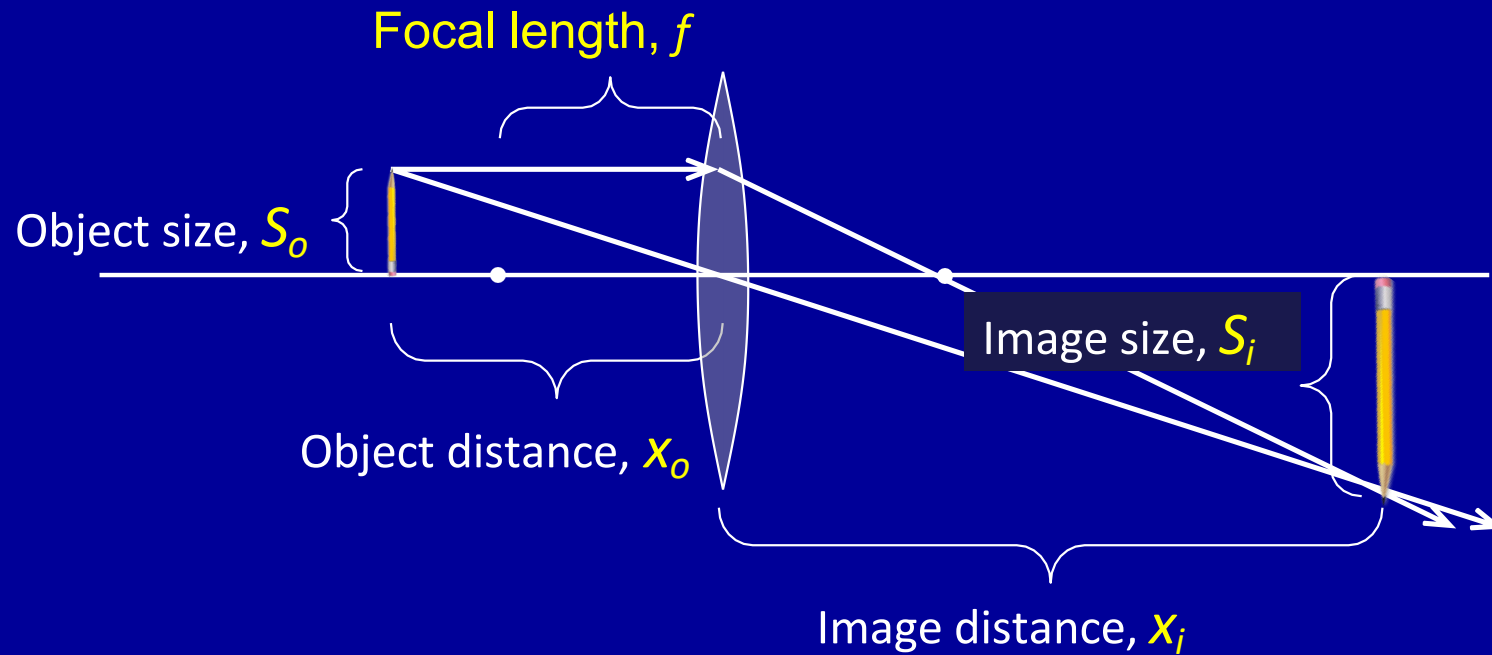
We can verify our result is consistent with the result from ray tracing. Ray tracing does not give an exact numeric answer, because we can't draw all our lines perfectly. But we can verify our answers.

Lens Equation: Magnification



- For objects and images above the axis, S is **positive**, for those below the axis, S is **negative**.

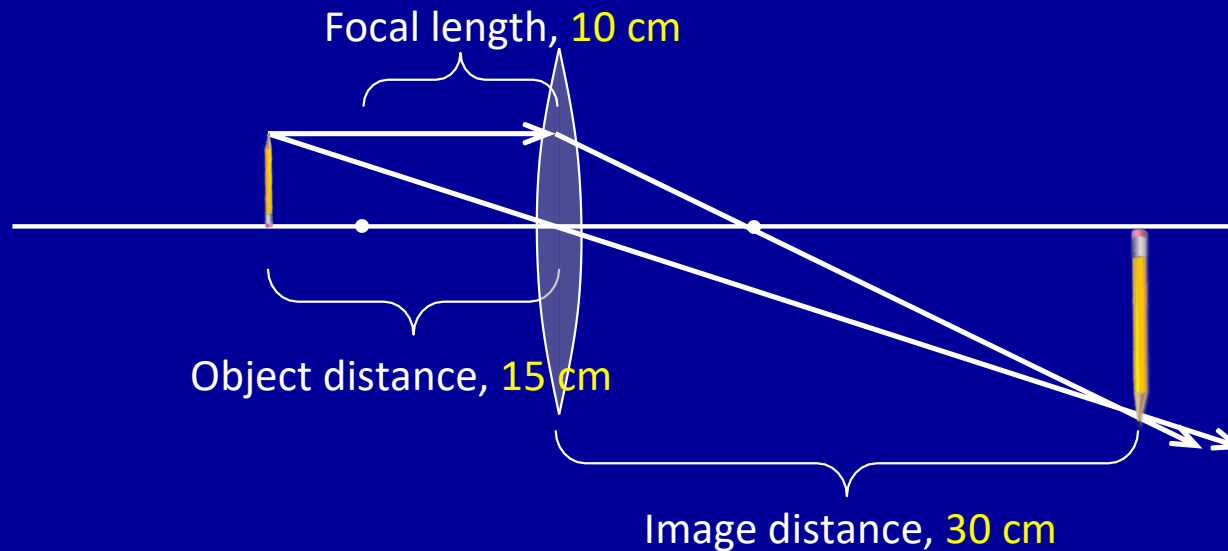
Magnification



$$M = \frac{S_i}{S_o} = \ominus \frac{x_i}{x_o}$$

Magnification is indicated by the letter M, and is negative for an image inverted with respect to the object, and positive for an image that is not inverted

Magnification



$$M = -\frac{x_i}{x_o} = -\frac{30 \text{ cm}}{15 \text{ cm}} = -2$$

Magnification of -2 means the image is twice the size of the object and inverted.

Lens Equation Example

- **Given:**

- $f = 10$ cm
- Object is 5 cm to left of lens center: $x_o = 5$

- **Find:**

- Where is image and is it real or virtual?

- **Solve** equation for x_i :

- Substitute numbers for letters
- Subtract $1/5$ from both sides
- Arithmetic on calculator
- Multiply by $x_i/0.1$

$$\frac{1}{x_o} + \frac{1}{x_i} = \frac{1}{f}$$

$$\frac{1}{5} + \frac{1}{x_i} = \frac{1}{10}$$

$$\frac{1}{x_i} = \frac{1}{10} - \frac{1}{5}$$

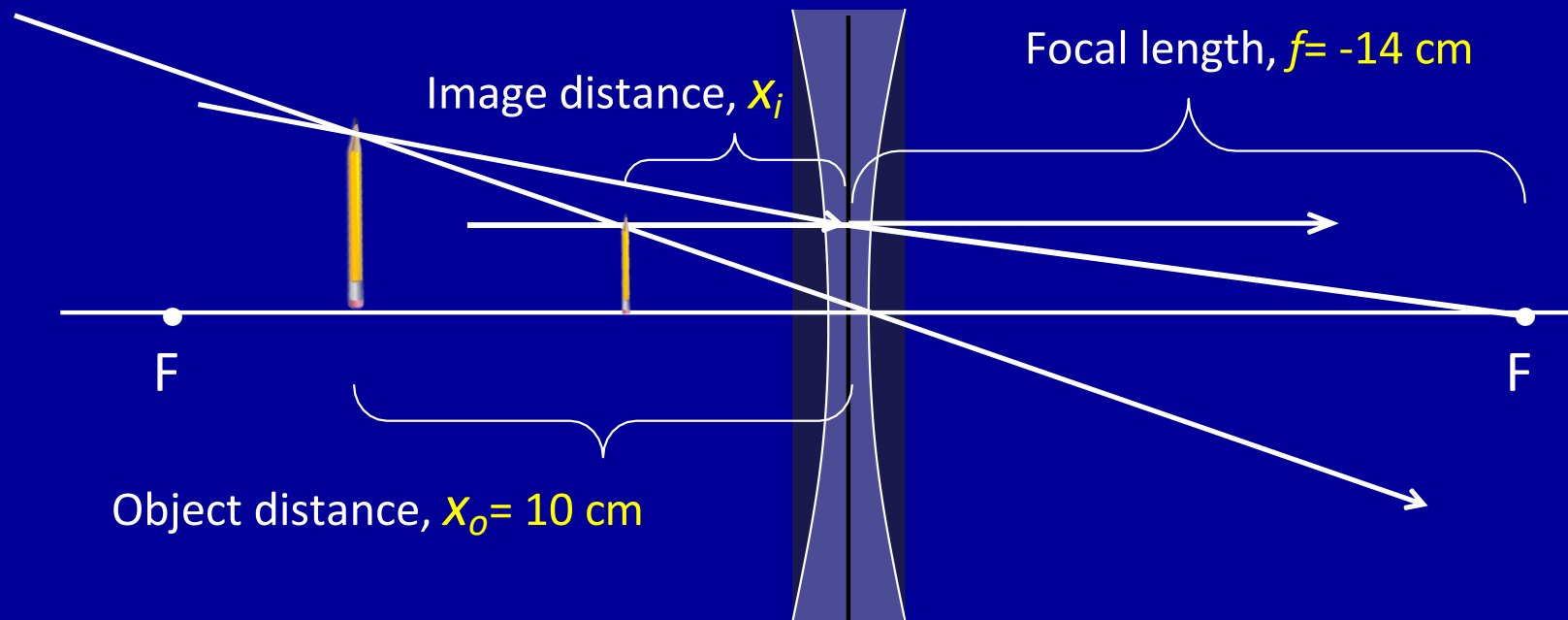
$$\frac{1}{x_i} = -0.1$$

$$x_i = -\frac{1}{0.1} = -10 \text{ cm}$$

Image is 10 cm to the **left** of the lens and **virtual** because x_i is negative

Lens Equation Examples

- Magnification
- Diverging lens



The “Power” of a Lens: Diopters

- Definition of P in terms of f is

$$P \text{ (in diopters)} = \frac{1}{f \text{ (in meters)}}$$

- Meaning of P
 - P is a measure of the ray bending power of the lens
 - Your eyeglass or contact lens prescription is usually given in diopters (P)
- P is positive for converging lenses (positive f) and negative for diverging lenses (negative f)

Clicker Question

- The power of the right lens in my glasses is -2.0 diopters. What is the focal length?

- A. 1 meter
- B. -1 meter
- C. 0.5 meters
- D. -0.5 meters

$$-2.0 \text{ diopters} = 1/f$$

$$f = 1/(-2.0 \text{ diopters})$$

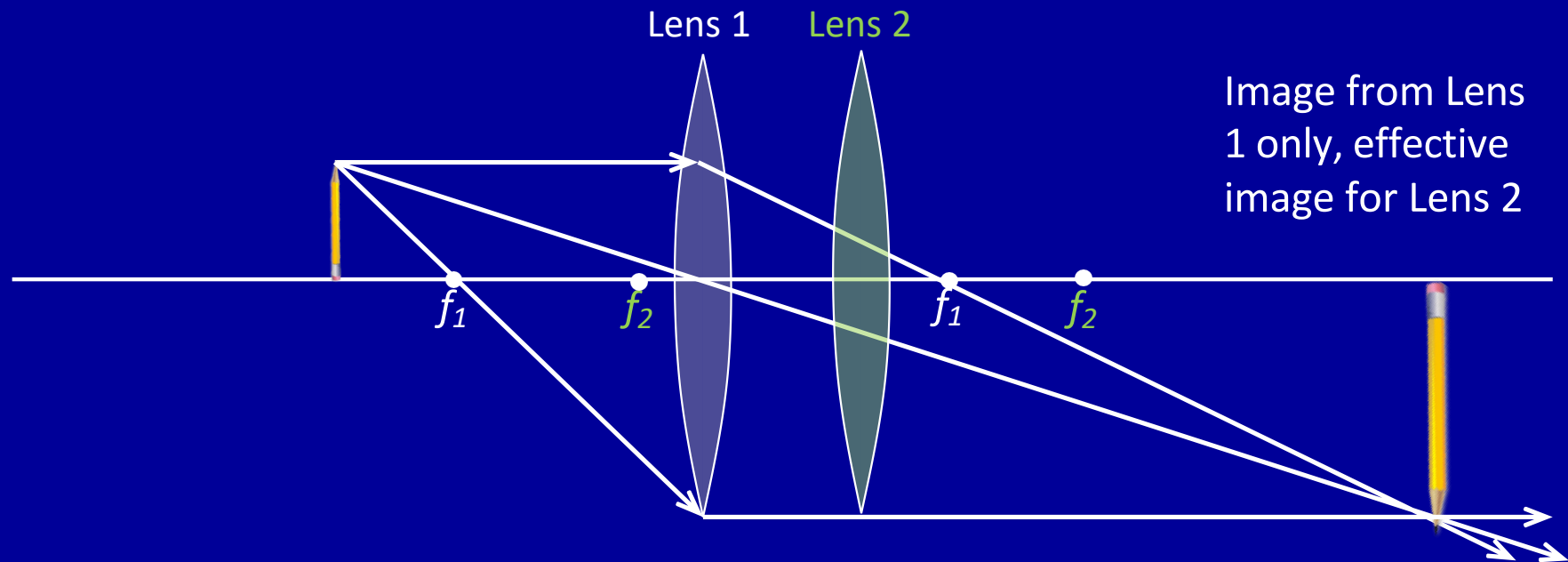
$$f = -0.5 \text{ meters}$$

$$P \text{ (in diopters)} = \frac{1}{f \text{ (in meters)}}$$

Combining Lenses: Using Diopters

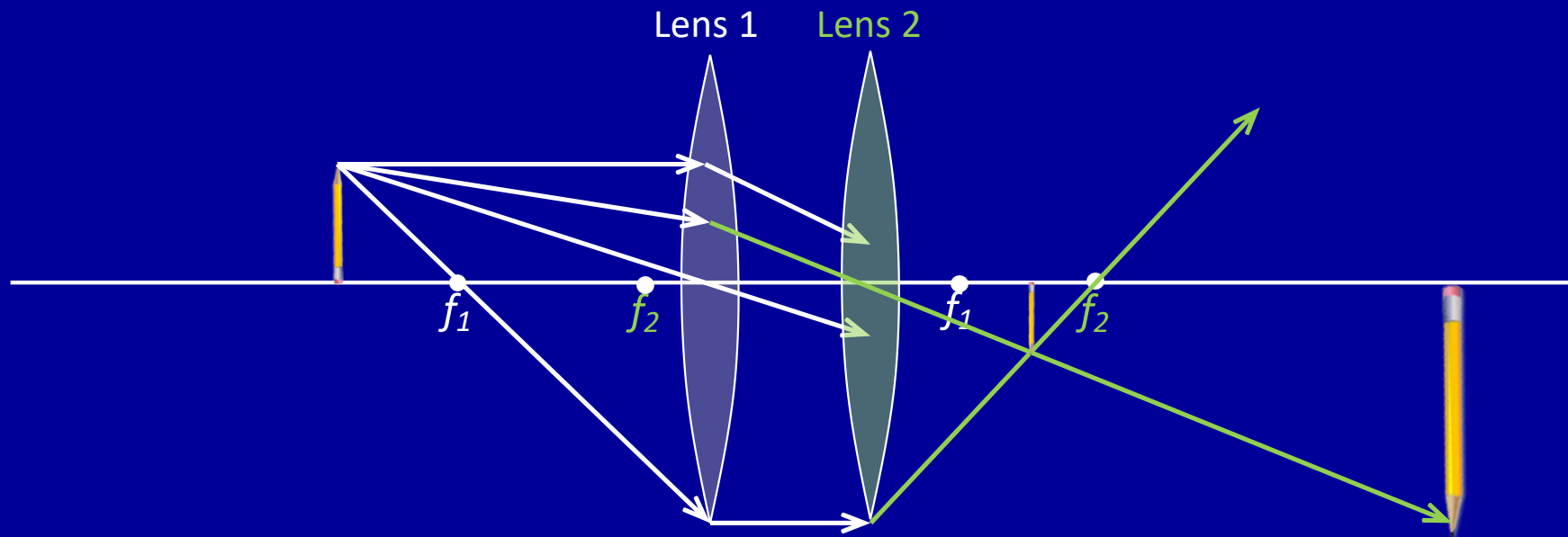
- Add together the powers of each of the two lenses to get the power of the combined lens
- Example:
 - o Lens A has $f_A = 0.5 \text{ m}$,
 - o Lens B has $f_B = -1 \text{ m}$.
 - o What is the power of each lens?
 - o What is the power of the combined lens when the two lenses are held together?
 - o What is the focal length of the combined lens?
- Power of lens A is 2 diopters
- Power of lens B is -1 diopters
- Power of combined lenses is $2 - 1 \text{ diopters} = 1 \text{ diopter}$
- Focal length of combined lens is $f = 1 \text{ meter}$
- This is only for THIN LENSES held so they are touching each other!

Multiple Lenses



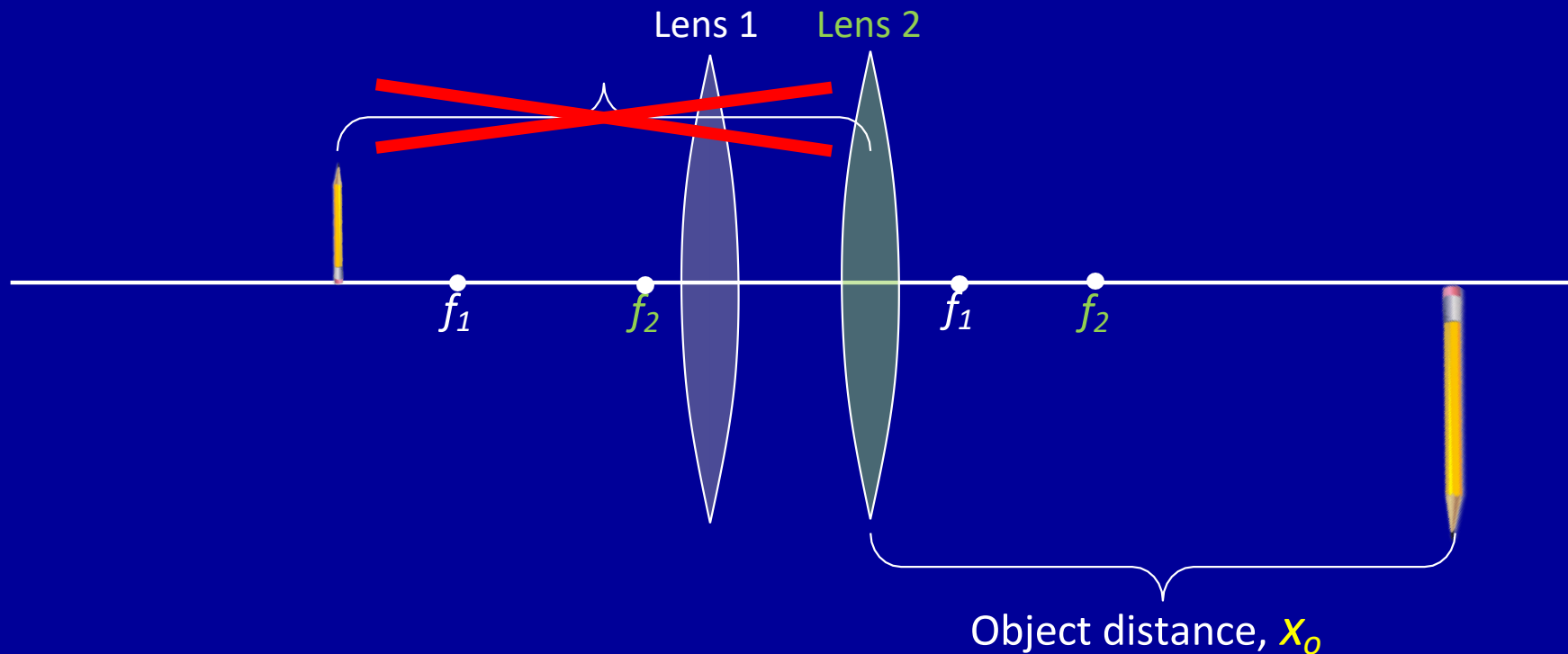
If we add a second lens, we can find the image produced by the combination of lenses by using the image from Lens 1 as an effective image for Lens 2

Multiple Lenses



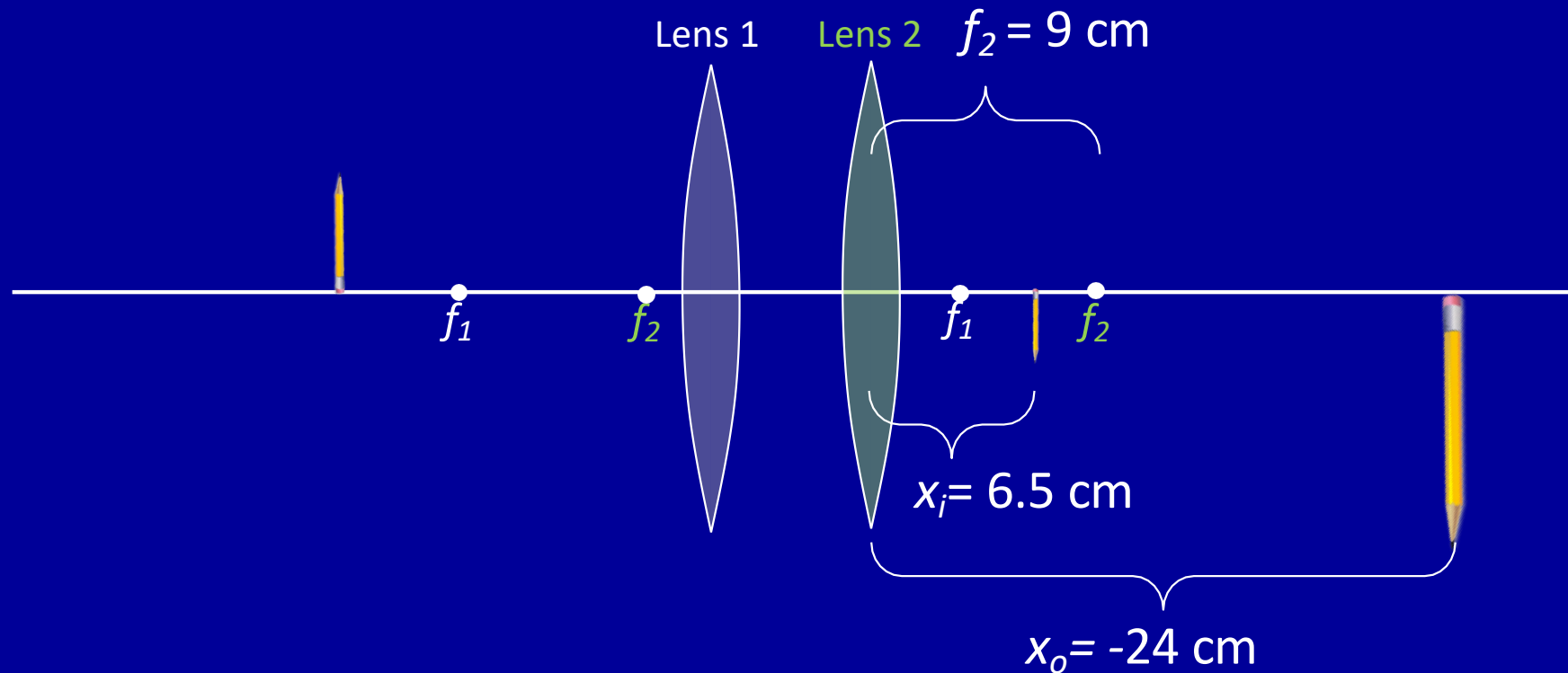
- We know where rays from the original object have to hit Lens 2 because we know where the image is. We can use this to find the special rays for Lens 2, and the final image.
- This is complicated, but we can use the Lens Equation!

Multiple Lenses: Lens Equation



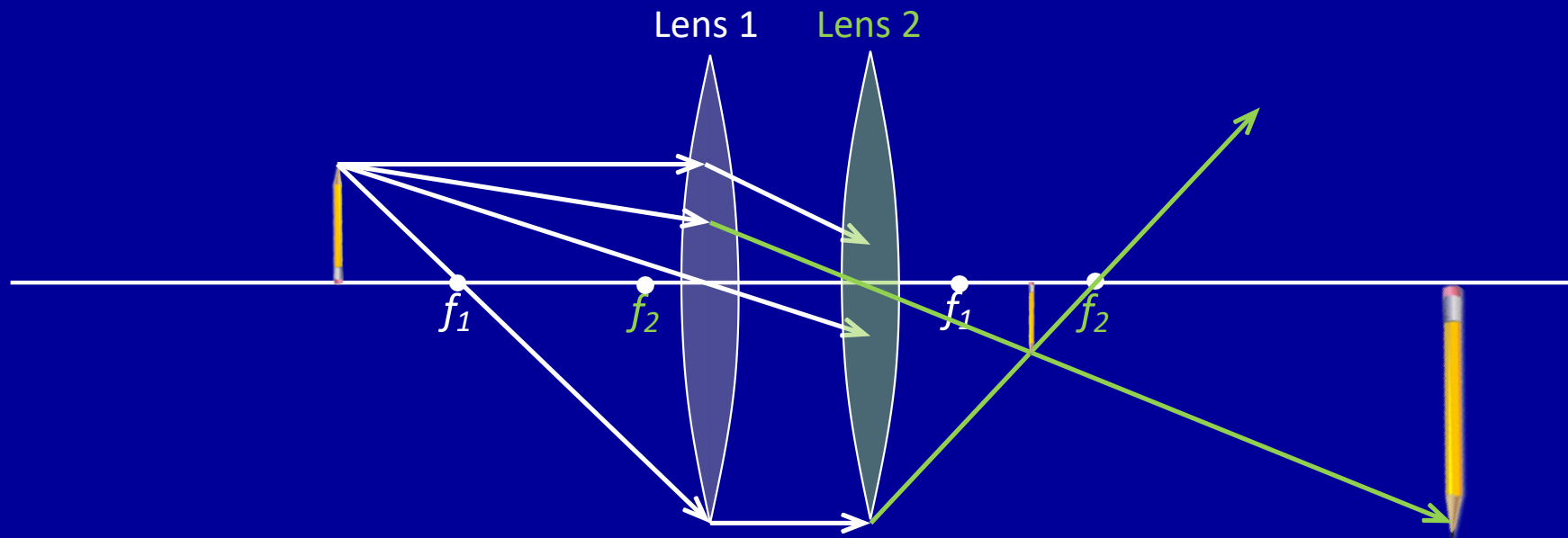
The trick to using the image from Lens 1 as an effective object with the lens equation is knowing what the object distance is for Lens 2. Note that it will be **negative**, because it is to the **right** of Lens 2.

Multiple Lenses: Lens Equation



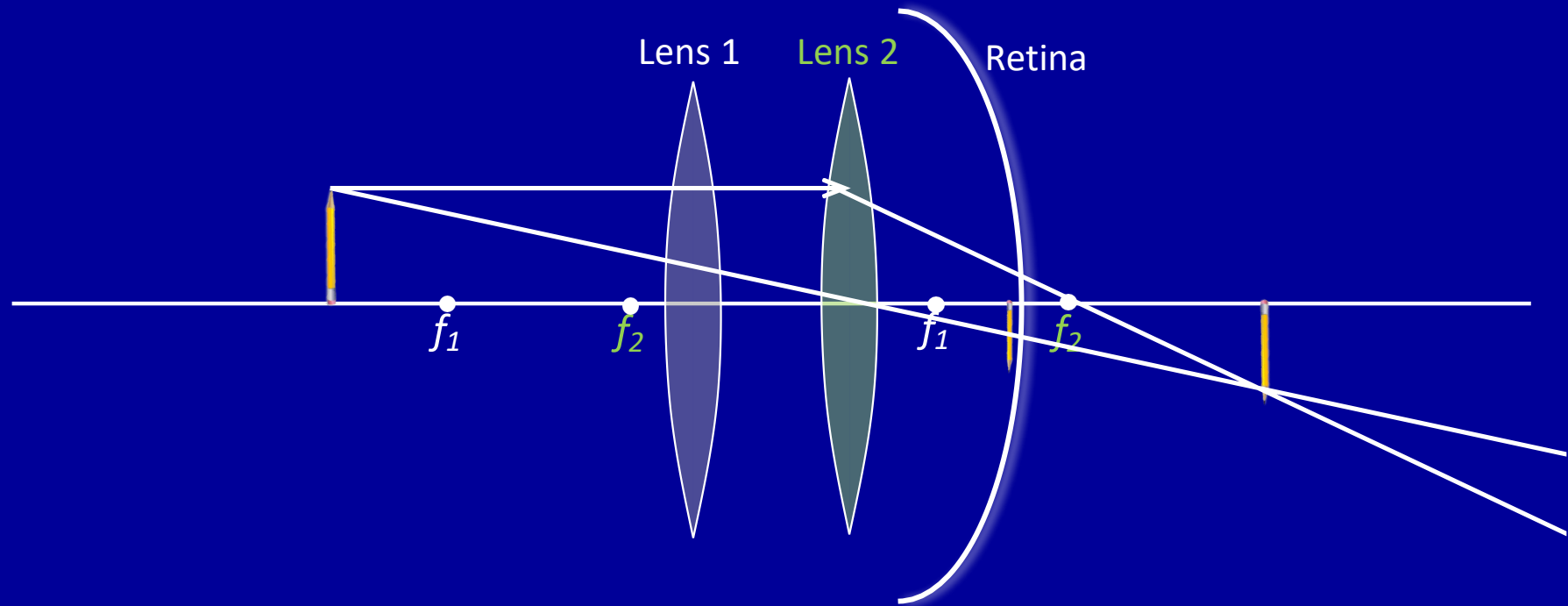
Let's add some numbers and see how it works.

Multiple Lenses



- The lens equation is consistent with our ray tracing result!

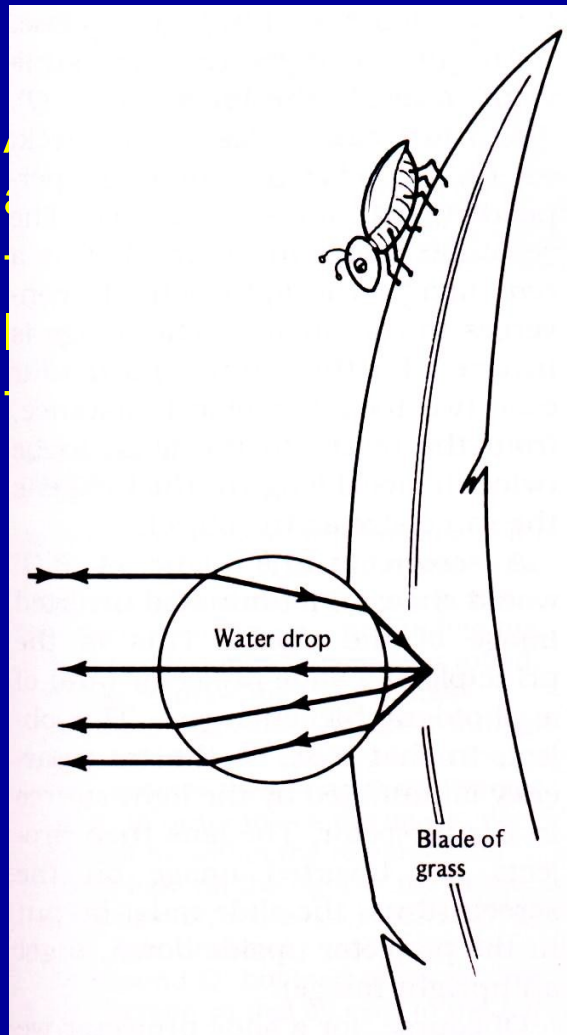
Corrective Lenses



- If you have 20/20 vision, the lens of your eye (Lens 2) produces an image correctly on your retina. Here it produces an image behind the retina. This looks blurry.
- If we add a lens in front, like glasses or contacts, the combination of the two lenses will produce an image correctly located at your retina

Ball Lens Retroreflector

- A
- S
- I
- T

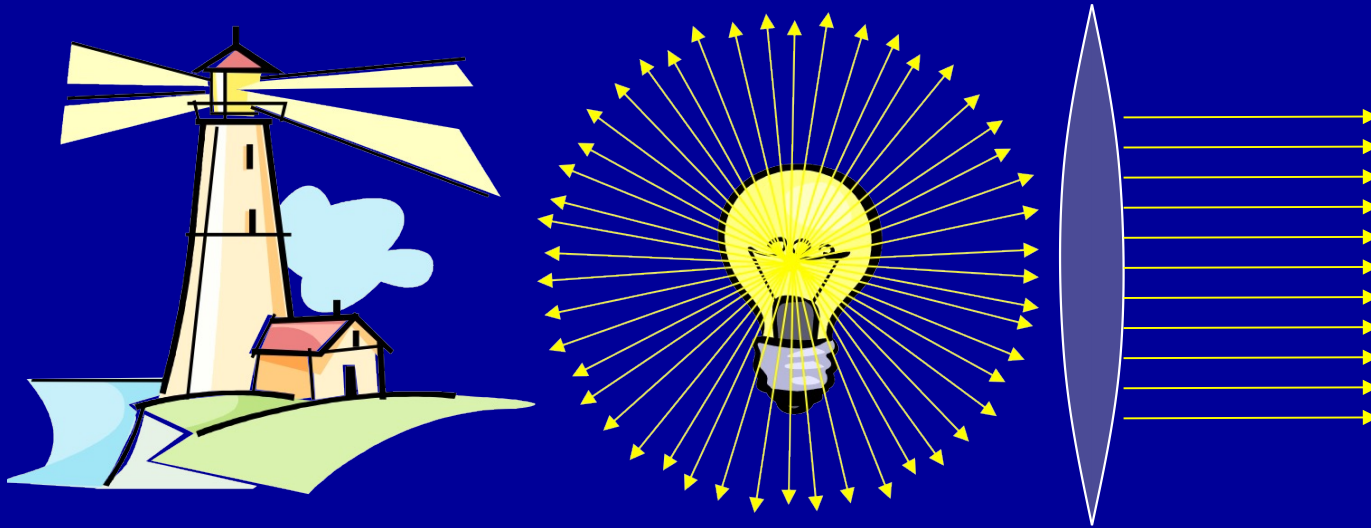


droplet on a blade of grass,
erging lens.
t the surface of the glass, so
ass and returns the way it came
oreflector

Safety Applications

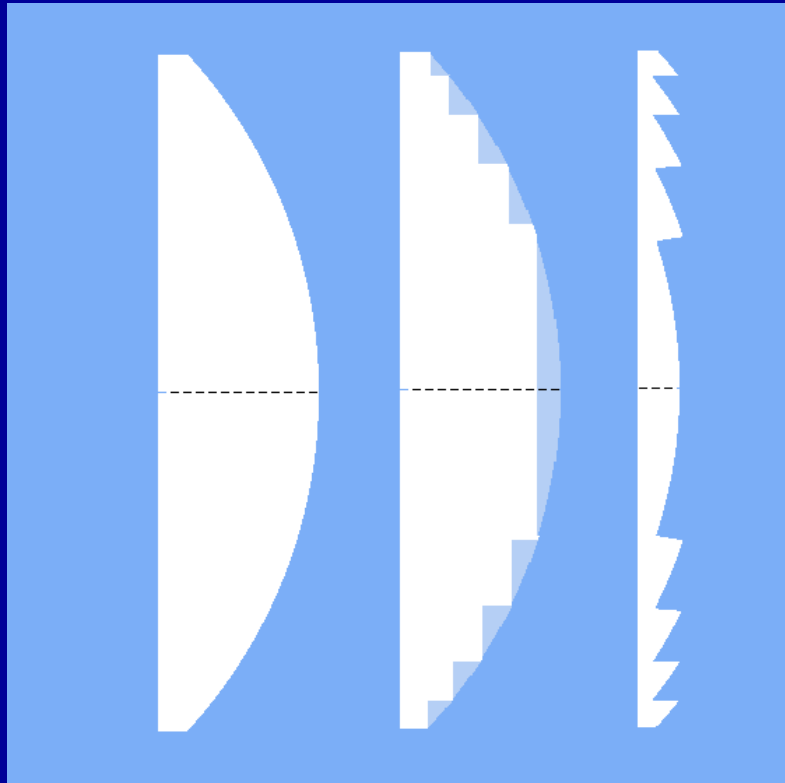


Fresnel Lens



A lighthouse wants to collect as much light as possible (large diameter lens) and send it out in a beam. This means the lens must have a short focal length (thick lens), so it can be placed close to the light source. This makes the lens very heavy.

Fresnel Lens



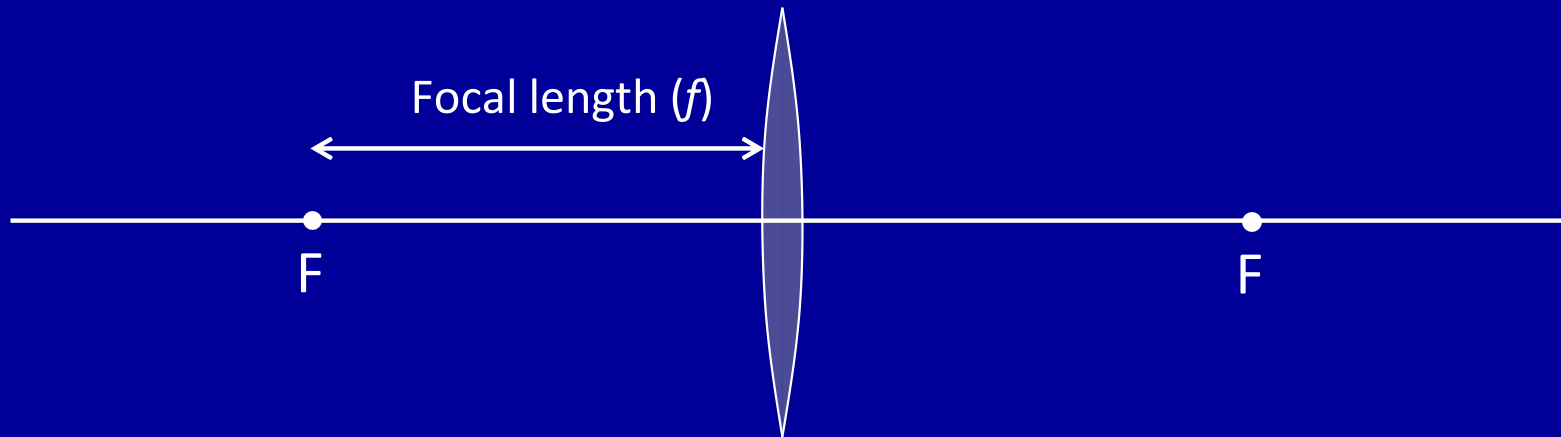
Fresnel, a French physicist, had the idea that because refraction occurs at the curved surface, the rest of the glass is unnecessary. If you remove the non-essential glass and flatten the remaining glass segments, you get a Fresnel lens. There are weird effects at the edges, but most applications do not require precision lenses, so this isn't a problem.

Fresnel Lenses: Applications



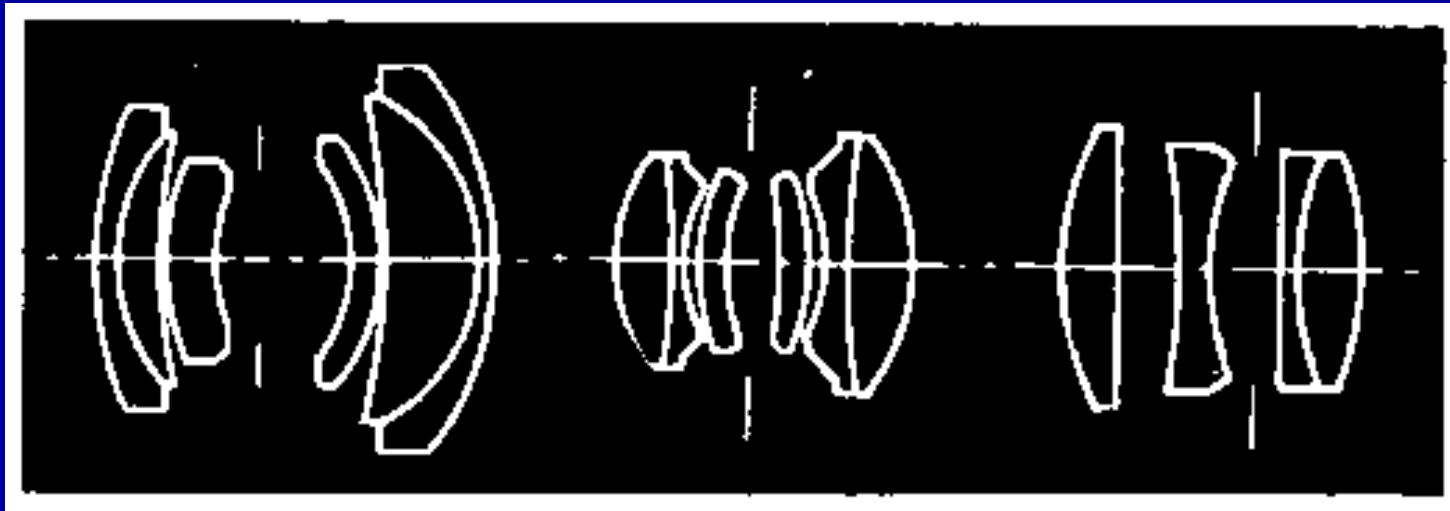
Focal Length

- Remember we defined the **focal length** for a lens



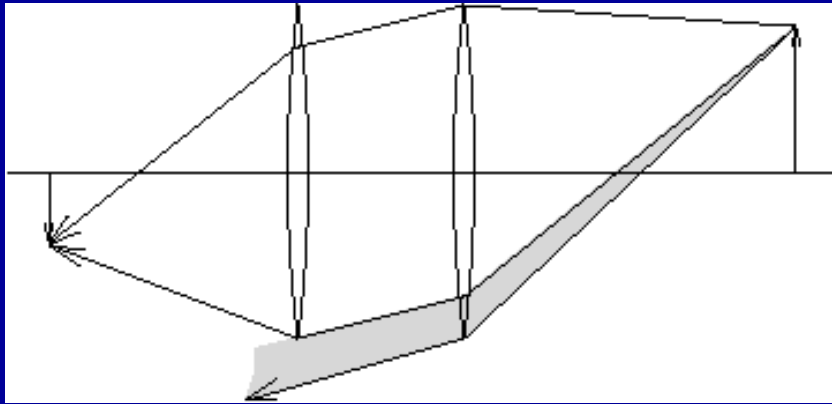
- We also defined the sign of f . The focal length, f , is defined as **positive** for converging lenses and **negative** for diverging lenses.

Lens systems



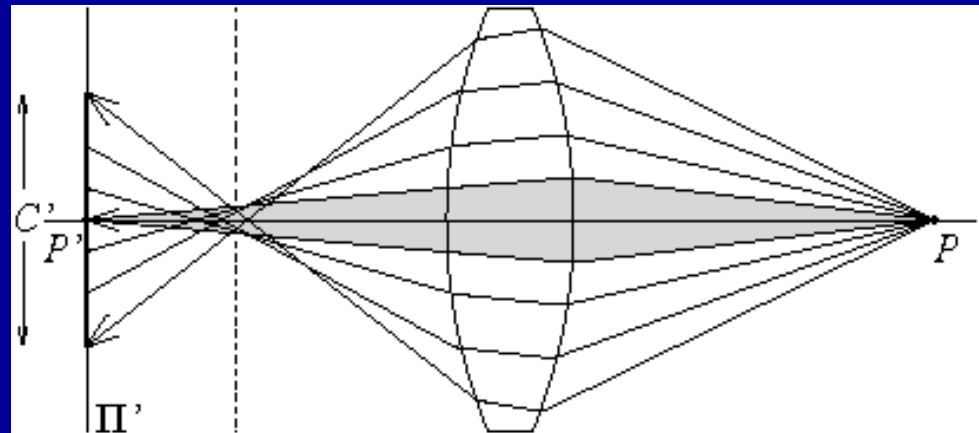
Lens Realities

Real lenses have a finite depth of field, and usually suffer from a variety of defects



vignetting

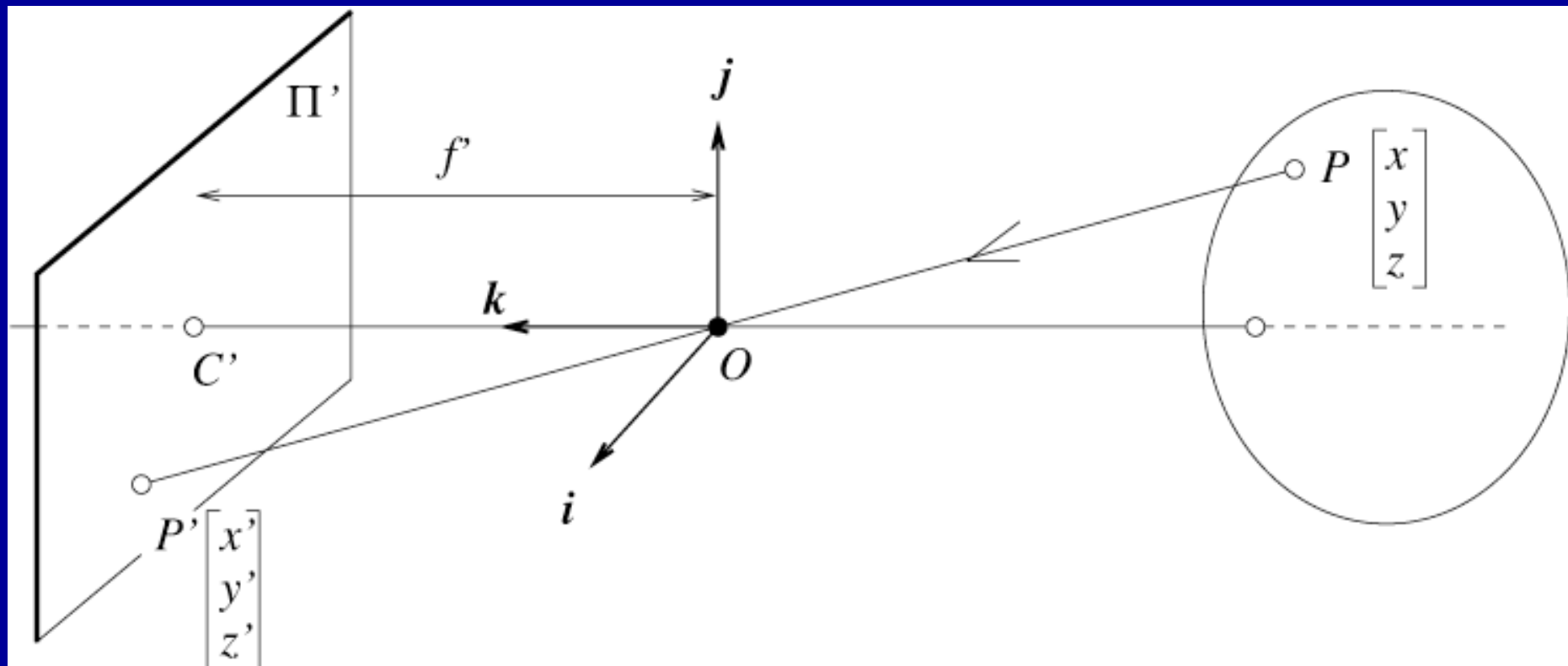
Spherical Aberration



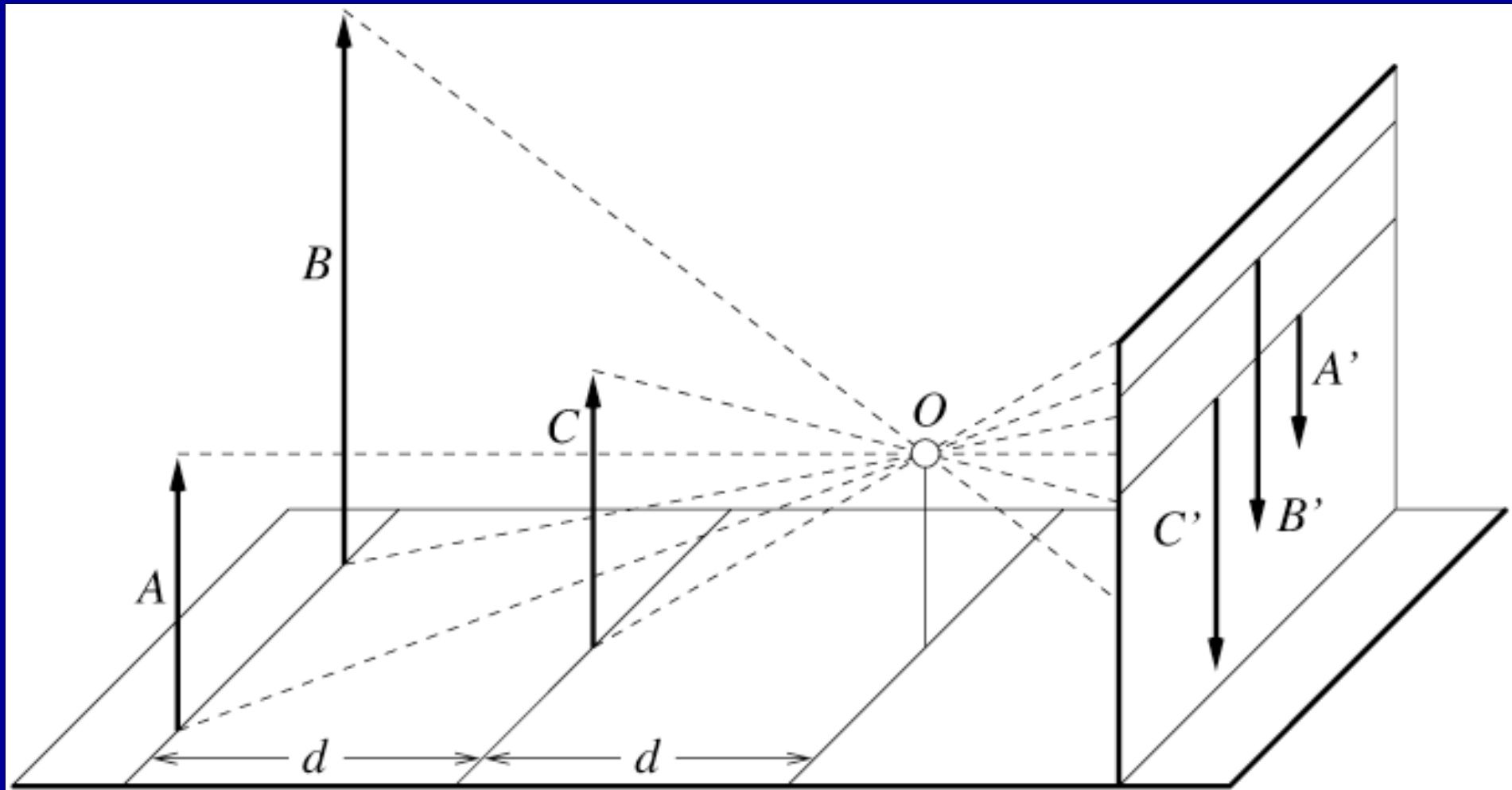
The equation of projection

- Cartesian coordinates:
 - We have, by similar triangles, that $(x, y, z) \rightarrow (f x/z, f y/z, -f)$
 - Ignore the third coordinate, and get

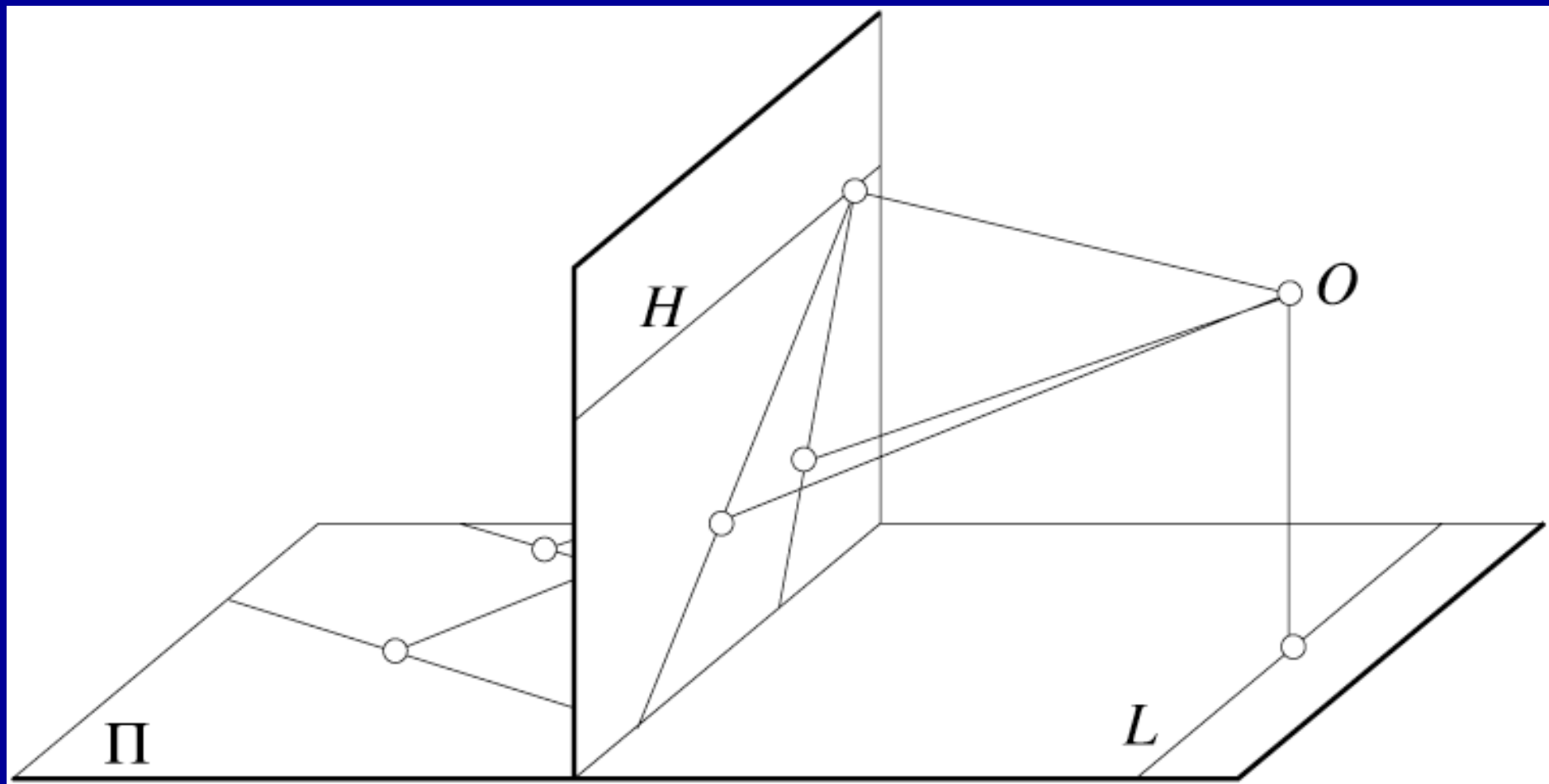
$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z} \right)$$



Distant objects are smaller

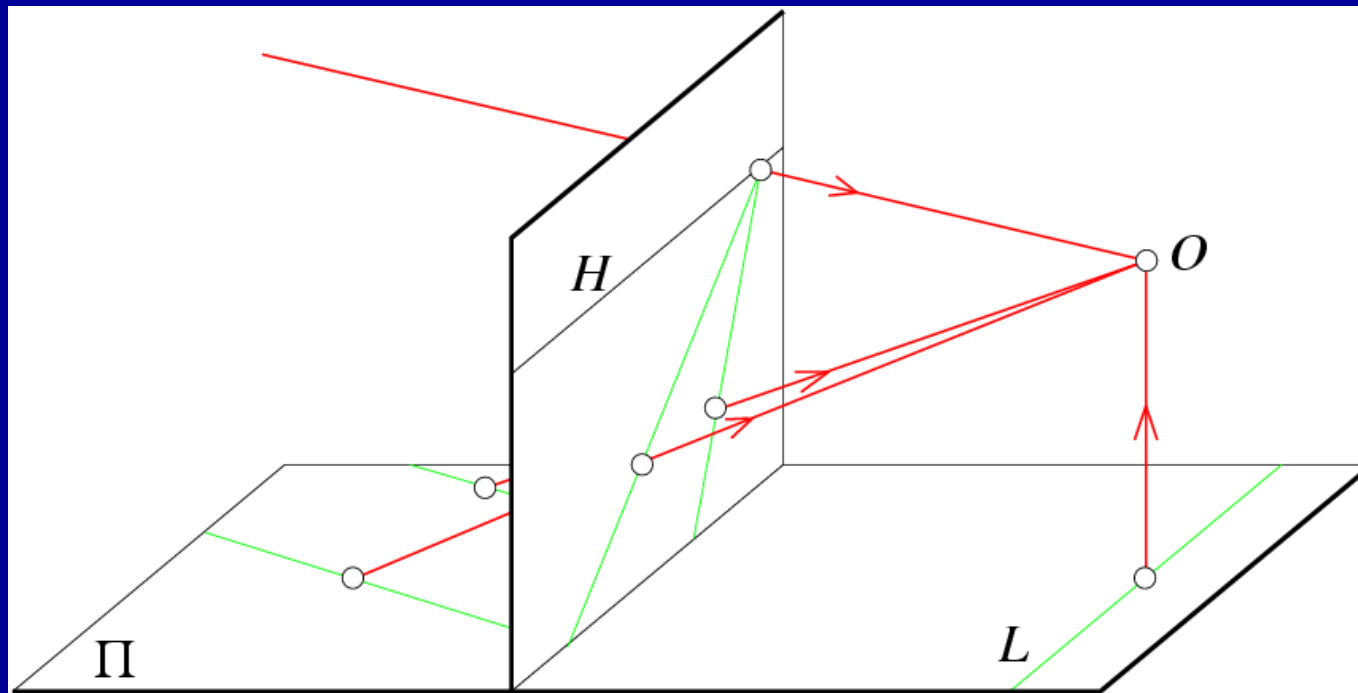


Parallel lines meet



Consequences: Parallel lines meet

- There exist vanishing points



Vanishing points

- each set of parallel lines (=direction) meets at a different point
 - The *vanishing point* for this direction
 - How would you show this?
- Sets of parallel lines on the same plane lead to *collinear* vanishing points.
 - The line is called the *horizon* for that plane

Implications For Perception*



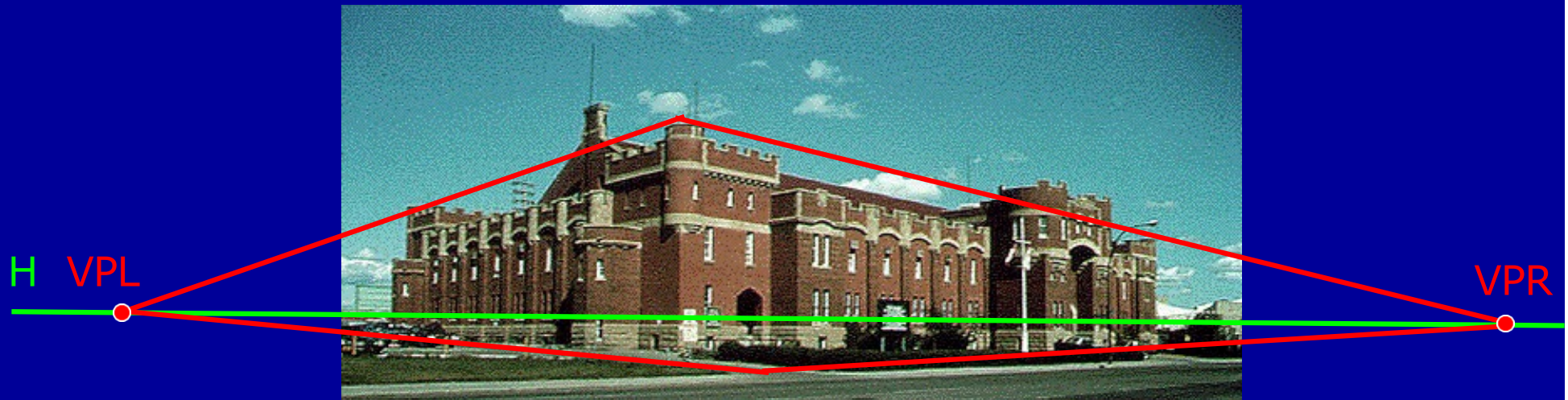
Same size things get smaller, we hardly notice...



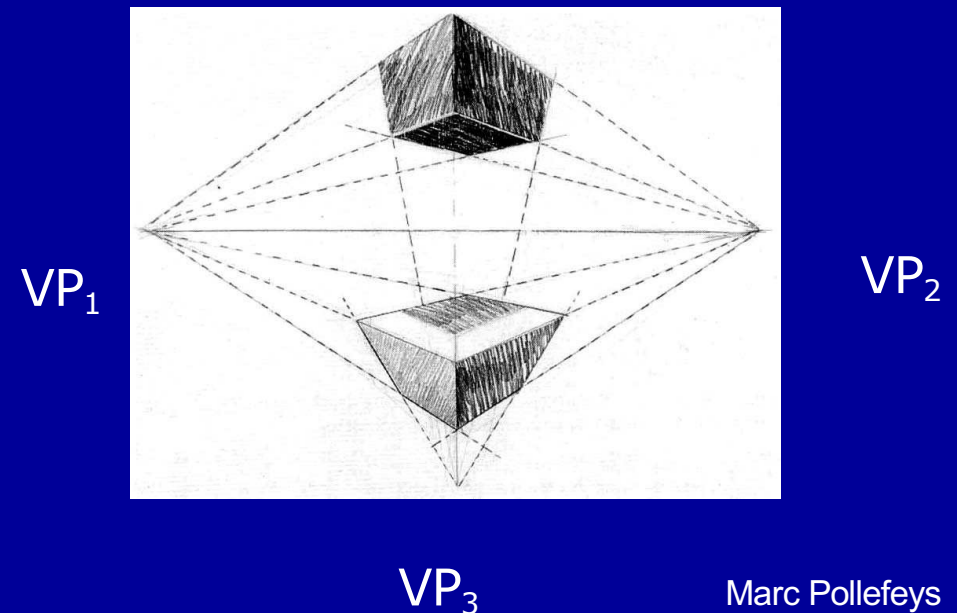
Parallel lines meet at a point...

* A Cartoon Epistemology: <http://cns-alumni.bu.edu/~slehar/cartoonepist/cartoonepist.html>

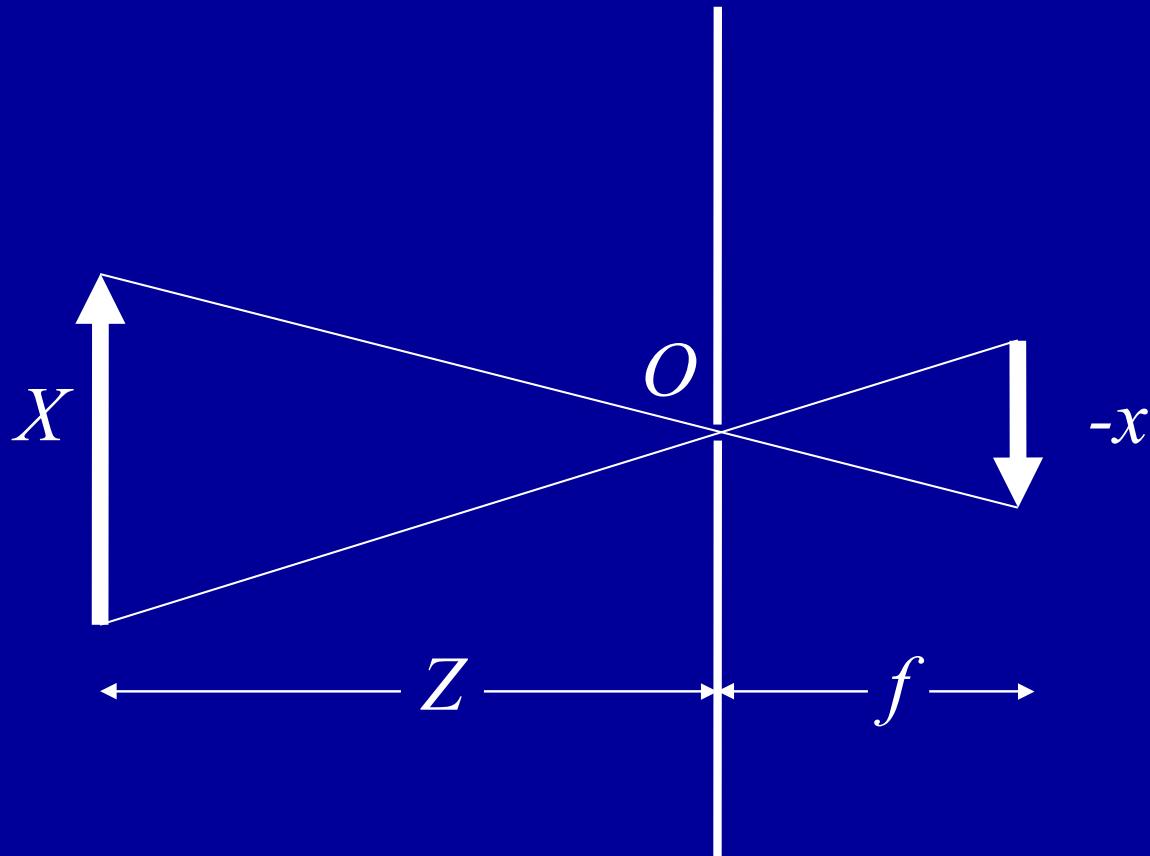
Vanishing points



Different directions correspond
to different vanishing points

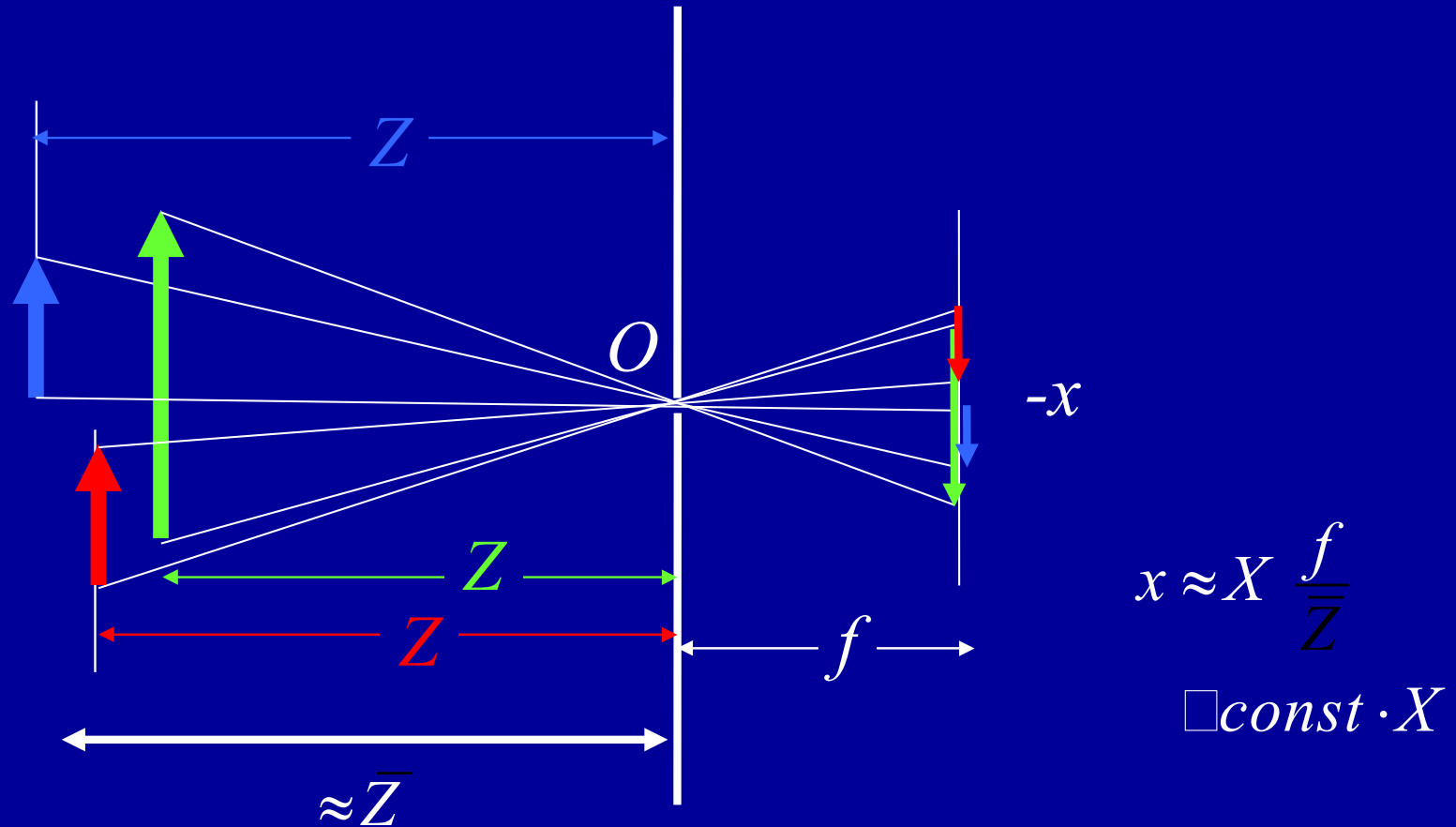


Perspective Projection

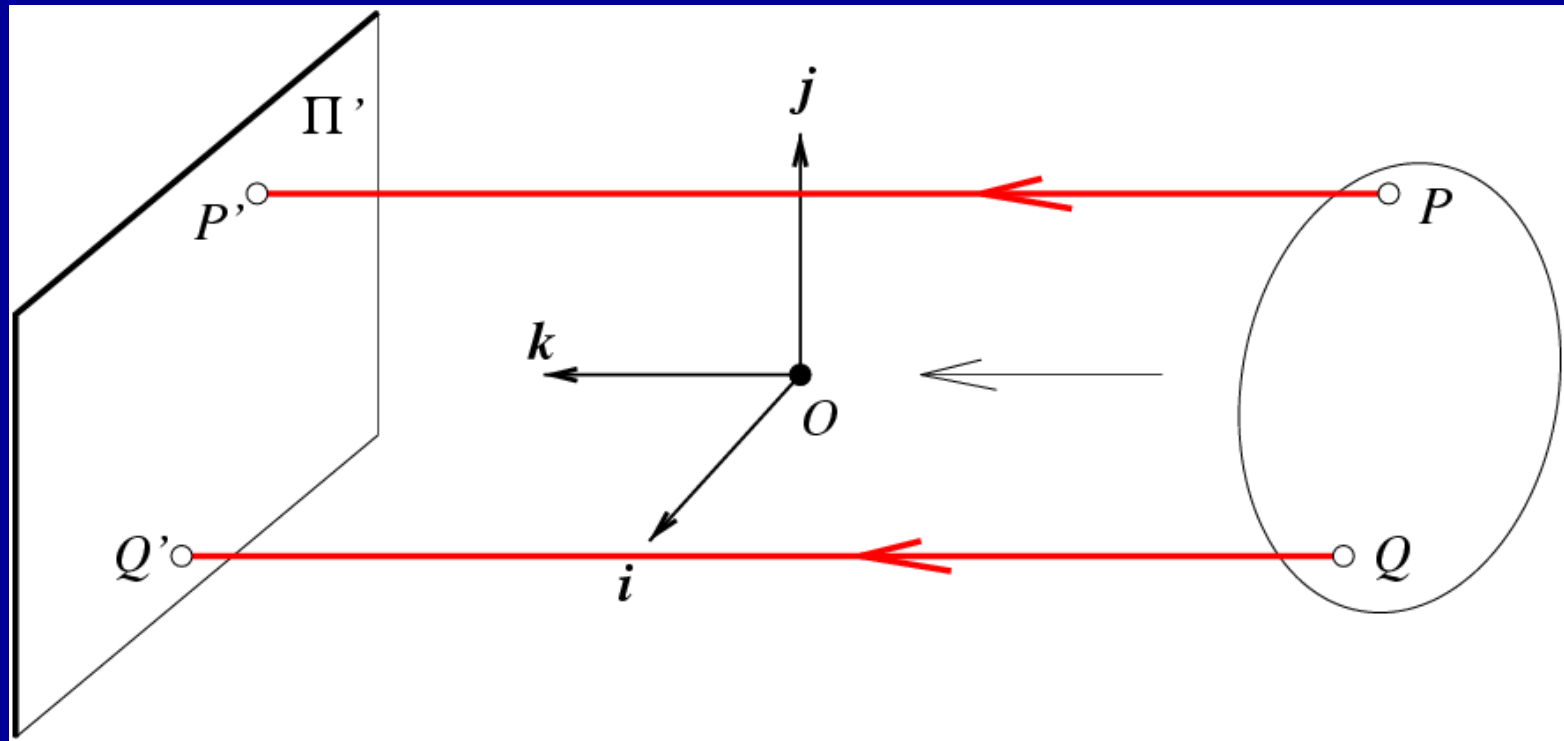


$$x \square X \frac{f}{Z}$$

Weak Perspective Projection



Generalization of Orthographic Projection



$$\begin{cases} X \approx x \\ Y \approx y \end{cases}$$

When the camera is at a
(roughly constant) distance
from the scene, take $m=1$.

Pictorial Comparison

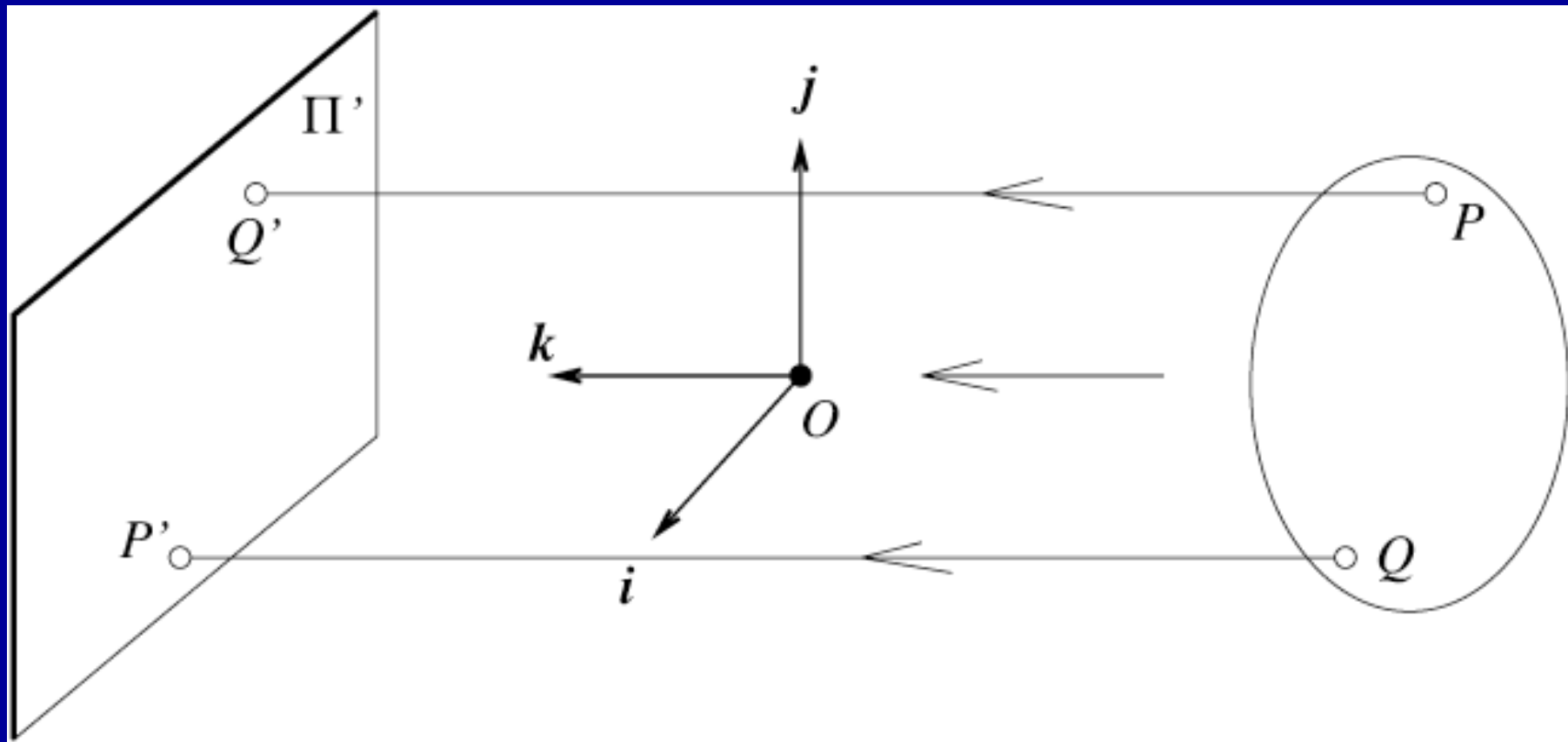
Weak perspective



Perspective



Orthographic projection



$$u \sqsubset x$$

$$v \sqsubset y$$

Weak perspective

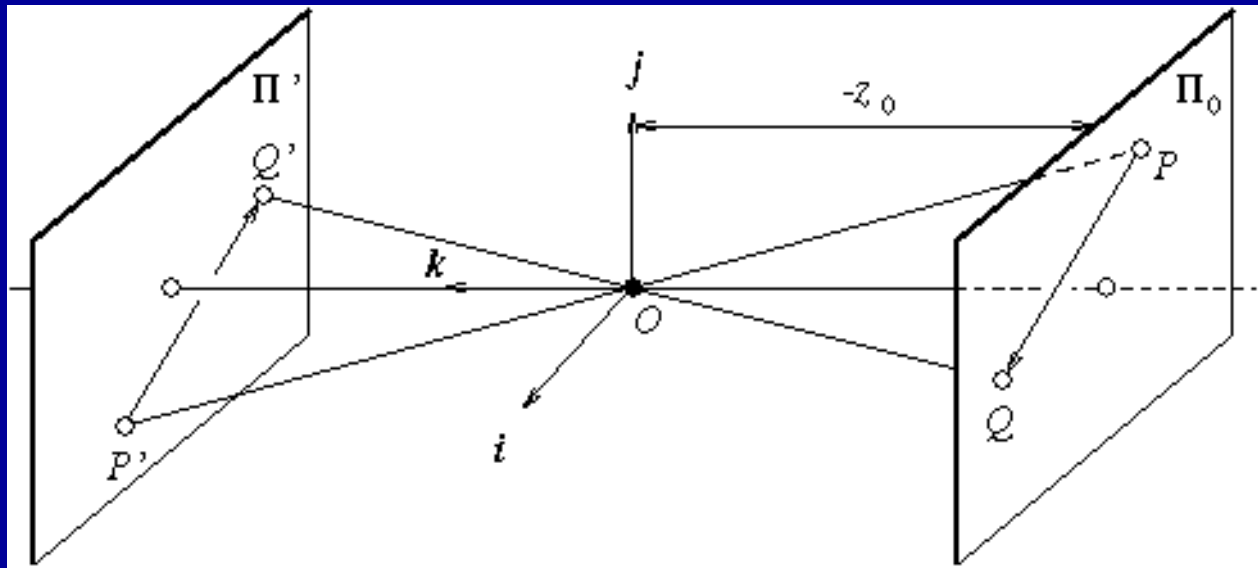
- Issue

- perspective effects, but not over the scale of individual objects
- collect points into a group at about the same depth, then divide each point by the depth of its group
- Adv: easy
- Disadv: wrong

$$u \approx Tx$$

$$v \approx Ty$$

$$T \approx f / Z$$



Other Models

- The *affine camera* is a generalization of weak perspective.
- The *projective camera* is a generalization of the perspective camera.
- Both have the advantage of being linear models on real and projective spaces, respectively.

Camera calibration

- Issues:
 - what are intrinsic parameters of the camera?
 - what is the camera matrix? (intrinsic+extrinsic)
- General strategy:
 - view calibration object
 - identify image points
 - obtain camera matrix by minimizing error
 - obtain intrinsic parameters from camera matrix
- Error minimization:
 - Linear least squares
 - easy problem numerically
 - solution can be rather bad
 - Minimize image distance
 - more difficult numerical problem
 - solution usually rather good, but can be hard to find
 - start with linear least squares
 - Numerical scaling is an issue